On the Ultrametric Unconscious, Quantum Consciousness, and Mental Spaces

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Abstract

In this thesis we attempt to model ideas of Ignacio Matte-Blanco and his ideas relating the unconscious to symmetry and consciousness to asymmetry. We attempt to model the unconscious by means of the p-Adic ultrametric. We then attempt to model consciousness by the 'collapse' of a quantum like superposition of propositional "logic waves". Lastly we develop ideas of a "mental space" which we give a topology based on propositional logic, build simplical complexes of logical expressions, and consider logical analogs to notions in algebraic topology such as homotopy and homology.

Contents

I would like to thank: ALGANT, Wolfram Alpha, Stack Exchange, Wikipedia, NCS Bar, and The Great Green Arkleseizure

Introduction

Recently there has been a large movement to find a mathematical model of the mind. Much of the progress made in developing a model has been on the substrate level, such as that of neuronal interaction, modeled for example, by neural networks in which the brain is a 'Black Box' in which, through training, particulars of Brain functionality is reached. This methodology allows for simulating various Brain functions. However, as in a computer simulation, there is a problem with the neural networks ability to *distinguish* itself from its environment and be conscious of its disassociation with it.

We approach the problem in a different manner and investigate the possibility that consciousness is a natural effect, built into the laws of nature; while consciousness may evolve, and this author believes it does, we investigate the possibility that initial spark of consciousness may lie in nature's ability to create asymmetries from symmetries.

We look at abstract mathematics, logic, quantum mechanics, and work done by Psychoanalyst Ignacio Matte-Blanco to build our model of the unconscious, conscious, as well as a topology of the mind.

This thesis is divided into 4 parts (sections):

Part 0: Math Basics. In this section we list 'basic' ideas which we develop on later in the thesis. The thesis has been structured such that notions which are common to mathematics; that is, can be found in off the shelf graduate or undergraduate text books or in online papers which give expositions on various topics in mathematics.

Part 1: Mathematical Constructions. In this part we develop the tools which will be used in the developing of the rest of the thesis work. The mathematical constructions found here were developed from basic ideas given in part 0 and may or may not be found in other works as they were developed specifically for the latter parts of the thesis; hence there will not be much in terms of references except to that which the author has found useful in the development of these mathematical constructions.

The main ideas of this section are as follows: The first part **??** consists of ways of representing logical expressions. The idea of partitioning logical expressions become extremely useful in the later development of a 'ProLT' or

a 'Propositional Logical Topology **??**. The logical correspondence matrices **??** have a multitude of uses which include being abstracted into representing larger logical expressions as tensors as well as, having a linear property, allow for the bra-ket representation which is native to quantum mechanics **??**. We then, using the found correspondence matrices, build a quantum like logic; a combination of logic and quantum mechanical principles such as superposition and measurement **??**.

In section 2 we explore the 'Mind'. We start by developing definitions in which we associate the mind to a topology, specifically topologies for the unconscious and conscious. As we treat the mind as a 'mental space' which consists of topologies (for the unconscious and conscious), we define how notions such as 'concepts' could be sets of data in these different topologies. We then model a particular idea of Matte-Blanco, such as his idea of the 'unfolding function' which can transform elements of the unconscious to that of the conscious. We model this function by what we call the 'distinguishing operator ∂'.

We go on to model the ideas of Matte-Blanco in which we obtain a symmetric topological space, or that in which each set is either equal to or indistinguishable from every other set. We employ the use of the Ultrametric which gives rise to the indiscrete topology **??**. We are able to create a space in which sets (concepts) become *indistinguishable* from one another. This gives us a symmetry in that one can not distinguish concepts hence is a possible model for the symmetric unconscious.

We then attempt to model consciousness. A motivation for our choice of using quantum mechanics for this model is inspired quantum field theories treatment of physical particles as excited states of an underlying physical field; that actual distinguishable particle/antiparticle pairs are created and the individual particles can be measured and trapped. In a sense this natural phenomenon is a breaking of symmetry as matter is produced from energy and has the ability to be 'measured'. While this 'measurement' is a seeming paradox itself, we apply a measurement on a 'symmetrical logical expression'. Then by employing phenomenon such as quantum collapse (but a logical version of it) we are able to obtain a collapse of symmetric logical expressions to an asymmetric logical expression.

It may be the case that there is a naturally occurring 'collapse' of some kind of underlying 'mental field' (possibly physical such as electromagnetic

brainwaves) which creates the Brain and hence Mind's ability to distinguish, even if just on a very rudimentary level. Thus by translating this possibility in to a simplistic axiomatic system such as propositional logic we might have the ability to observe what is happening due to this collapsing effect in a mathematical way.

The possibility for observing quantum like effects in the Brain or Mind via propositional logic gives the possibility for examining structures that can be created with mathematics. This is the content of section 3. Our aim is to establish how logic can be used to build 'higher order' structures such as simplicial complexes of concepts related by logical expressions.

We start by developing a Topology based on propositional logic and build more familiar objects (such as presheaves) so as to eventually give this 'Logical Topology (ProLT) the ability to be manipulated via algebra. We continue on this path and develop notions familiar to Algebraic Topology (but with a logical twist) such as Logical Exact Sequences and Logical Simplicial Complexes. We examine other fundamental ideas in Algebraic Topology such as Homotopy and finally Homology and show that a logical analog of these notions work.

0 Mathematical Basics

Motivation: In our pursuit of understanding the "Mind" we must first start with some mathematical basics.

In this section (chapter 0) we will deal with the mathematical ideas that are 'general' knowledge, meaning that most or all of these concepts can be found in a mathematics textbook aimed at a student.

0.1 Logic

Motivation: In the study of logic one employs a set of axioms in order to define a system. Letting the mind be treated as the system and the axioms be those of (Aristotelian) logic we aim to ultimately express mental processes in terms of a logic.

We first must define some concepts.

Definition *(Logical Expression)*

A *logical expression* $\mathcal{L}$ is a declarative statement which either has a true or false value. The set of logical expressions in CNF (Conjunctive Normal Form) is written $\mathbb{L}$. ♠

Definition *(Atom)*

An *atom* or an *atomic formula* is a single element of a logical expression which contains no logical structure or connectives, or equivalently atoms are formulas which contain no strict sub-formulas. ♠

Definition *(Logical Negation)*

Let $L \in \mathbb{L}$ be a logical expression. The *logical negation* of L denoted $\neg L$ is the operation $\neg$ which gives L the opposite truth valuation. ♠

Definition *(Literal)*

A *literal* is an atomic formula (atom) or its negation. Thus literals have two forms, first either a *positive literal* which is an atom, or a *negative literal* which is just the negation of an atom. ♠

Definition *(Propositional Variable [Logical Operand])*

A *propositional variable* or *logical operand* is a variable in a logical expression which can be assigned a truth value. ♠

Definition *(Truth Values)*

The set of *truth values* written $\mathbb{T}$ is the set of values $\mathbb{T} = \{F, T\}$ where $F =$ False and $T =$ True. ♠

Definition *(Logical Operator)*

A *logical operator* is an operator (function) $f : (\mathbb{T}, \mathbb{T}) \to \mathbb{T}$. ♠

Definition *(Truth Table)*

A *truth table* is the set of maps over logical sentences. It is a two-dimensional array with $n + 1$ columns. The first n columns correspond to the inputs to the logical literals and the last column to the logical expression in which the truth values are assigned. The rows of the truth table are those of unique sets of truth assignments to logical literals. ♠

The following several definitions are of simple logical expressions (sentences). The corresponding truth tables are listed in figure 1.

Definition *(Valuation)*

A *truth valuation* is a map $\mathcal{V} : \mathbb{L} \to \mathbb{T}$ where $\mathbb{L}$ is the set of logical expressions and $\mathbb{T}$ is the set of truth values. ♠

Thus we are assigning truth values to each of the literals and each collection of truth value assignments is a particular valuation.

Definition *(True and False Valuation)*

A *true valuation* (resp. *false*) is a valuation in which each literal is assigned to be true (resp. false). We will denote this case by $\mathcal{V}_T$ (resp.$\mathcal{V}_F$). ♠

Example 0.1 *Let* $\mathbb{L} \ni \mathcal{L} = (X \wedge \neg Y) \vee (\neg X \wedge Y) = X \Leftrightarrow \neg Y$. *The true valuation* $\mathcal{V}(\mathcal{L}) = T$ *(True) is the set of truth values* $\{\{T, F\}, \{F, T\}\}$ *corresponding to the tuple* $\{\mathcal{V}(X), \mathcal{V}(Y)\}$.

X	Y	$X \wedge \neg Y \vee \neg X \wedge Y$
F	F	F
F	T	T
T	F	T
T	T	F

Thus we have that $\mathcal{V}_T = \{\{T, F\}, \{F, T\}\}$ *and say that* $\mathcal{V}(\mathcal{L}) = T$ *for the valuations* $\{\{T, F\}, \{F, T\}\}$. ♦

Definition *(Logical Conjunction)*

Let $X, Y \in \mathbb{L}$ be two logical expressions. The *Logical Conjunction*, commonly referred to as the *AND* operator, is denoted by the $\wedge$ symbol. ♠

Definition *(Logical Disjunction)*

Let $X, Y \in \mathbb{L}$ be two logical expressions. The *Logical Disjuction*, commonly referred to as the *OR* operator, is denoted by the $\vee$ symbol. ♠

Definition *(Logical Implication)*

Let $X, Y \in \mathbb{L}$ be two logical expressions. The *implication* operator gives the valuation corresponding to the truth table of $(\neg X \vee Y)$ that is $X \Rightarrow Y$. Sometimes implication is also written in the opposite way $\Leftarrow$ and we have for example $X \Rightarrow Y = Y \Leftarrow X$. ♠

It will become an indispensable fact that $X \Rightarrow Y = (\neg X \vee Y)$. This can be seen if one compares the corresponding truth tables in figure 1.

Definition *(If And Only If $\Leftrightarrow$)*

Let $X, Y \in \mathbb{L}$ be two logical expressions. The *if and only if* operator or *symmetric implication* operator is written $\Leftrightarrow$ and is defined by the logical expression $X \Leftrightarrow Y = (X \Rightarrow Y) \wedge (Y \Rightarrow X)$. ♠

Definition *(XOR Operator $\Updownarrow$)*

Let $X, Y \in \mathbb{L}$ be two logical expressions. The *XOR operator* is written $\Updownarrow$ and defined by the logical expression $X \Updownarrow Y = (X \wedge \neg Y) \vee (\neg X \wedge Y)$. ♠

X	Y	$\neg X$	$X \wedge Y$	$X \vee Y$	$X \Rightarrow Y$	$X \Leftarrow Y$	$X \Leftrightarrow Y$	$X \Updownarrow Y$
F	F	T	F	F	T	T	T	F
F	T	T	F	T	T	F	F	T
T	F	F	F	T	F	T	F	T
T	T	T	T	T	T	T	T	F

Figure 1: Logical expressions with operators $\neg, \wedge, \vee, \Rightarrow, \Leftarrow, \Leftrightarrow, \Updownarrow$

Definition *(Universal Set)*

A *universal set* is a set which contains all elements other than itself. ♠

Definition *(Tautology)*

A *tautology* is a Boolean sentence in which any set of assigned truth values for the literals result in the truth value of the Boolean sentence being true. ♠

Definition *(Boolean Algebra)*

A *boolean algebra* is be defined to be a set $B = \{a, b, \cdots\}$ with the following properties:

1) B has the operations $\wedge$ and $\vee$ which satisfy the idempotent laws:

$$a \wedge a = a \vee a = a$$

The commutative laws:

$$a \wedge b = b \wedge a \; a \vee b = b \vee a$$

The associative laws:

$$a \wedge (b \wedge c) = (a \wedge b) \wedge c \; a \vee (b \vee c) = (a \vee b) \vee c$$

2) The operations satisfy the absorption laws:

$$a \wedge (a \vee b) = a \vee (a \wedge b) = a$$

3) The operations are mutually distributive:

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)\ a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

4) B contains the empty set $\emptyset$ and the universal (universe) set I where:

$$\emptyset \wedge a = \emptyset\ \emptyset \vee a = a\ I \wedge a = a\ I \vee a = I$$

5) B has a unitary operation $a \rightarrow \neg a$ in which:

$$a \wedge \neg a = \emptyset\ a \vee \neg a = I$$

♠

0.2 Topology

Motivation: In mathematics we have the notion of a *Topology* to model spaces (i.e. Euclidean, and p-adic) to see how we might embed objects in these spaces as well as what properties such as continuity can be associated with them. To model a mental space we must find out how to embed objects such as *thoughts* and *concepts* (see section **??**) so that the manipulation of any of the objects which we want to model corresponds to the actual behavior observed in another space such as that of our physical universe. To start with we begin by identifying the space around points. This consists of, for each point, looking at the other points in the same set and the relationship between the various sets.

Definition *(Basic Neighborhood)*

Let X be a non-empty set, and suppose that for each $x \in X$ there exists a non-empty family of subsets of X denoted U_x that satisfy the following properties:

1. If $x \in U\ \forall U \in U_x$;

2. If $U \in U_x$ and $V \in U_x$ then $\exists W \in U_x$ such that $W \subset U \cap V$;

3. If $y \in U$ and $U \in U_x$ then $\exists V \in U_y$ with $V \subset U$

Each $U \in U_x$ is called a *basic neighborhood* of x. ♠

Example 0.2 *For a set* $X = \{3, 7, 11\}$, $U_3 = \{\{3\}, \{3, 7\}\}$, *and* $U_7 = \{\{7\}, \{7, 11\}\}$, *and* $U_{11} = \{\{7, 11\}\}$ *We have that* $\{3\}, \{3, 7\}$ *are basic neighborhoods of the point* $x \in X$ *when* $x = 3$. *For* $x = 7$ *the basic neighborhoods are* $\{7\}, \{7, 11\}$ *and for* $x = 11$ *it is* $\{7, 11\}$. ♦

Definition *(Basic Neighborhood System and Space)*

The collection $\{U_x | x \in X\}$ is called a *basic neighborhood system* for X. If B_X denotes a basic neighborhood system, the pair (X, B_X) is called a *basic neighborhood space.* ♠

Example 0.3 *Let* X *be a non-empty set and let* $U_x = \{\{x\}\} \forall x \in X$. $\mathcal{D}_X = \{U_x | x \in X\}$ *is called the discrete basic neighborhood system for* X. $(X, \mathcal{D}_X)$ *is called a discrete basic neighborhood space.* ♦

Example 0.4 *Let* X *be a non-empty set and let* $U_x = \{X\} \forall x \in X$. $\mathcal{I}_X = \{U_x | x \in X\}$ *is called the indiscrete basic neighborhood system for* X. $(X, \mathcal{I}_X)$ *is called a indiscrete basic neighborhood space.* ♦

The last two examples are significant as they show the separability and inseparability of points (respectively). The next example shows the indistinguishably of points in a space, such as that of the points in an ultrametric space (see **??**). This will become useful in other investigations of the ultrametric space and ultimately the unconscious!

We want to know whether or not one space can be morphed into another via a continuous map. This will help determine whether concepts and various logical expressions (which we will develop) can, and to what extent, be related and whether or not one expression can be manipulated to be another. We start with the definition for continuity via infinitesmals in $\mathbb{R}^n$

Definition *(Continuity)*

A function $f : \mathbb{R}^n \to \mathbb{R}^n$ is continuous at $x \in \mathbb{R}^n$ iff $\forall \epsilon > 0 \, \exists \, \delta > 0$ such that $f(B_\delta(a)) \subseteq B_\epsilon(f(a))$ where $B_r(x)$ is a ball in $\mathbb{R}^n$ with radius r centered at point x. ♠

If however for $r \in \mathbb{R}$ we take the balls $\{B_r(x)\} = \{(x - r, x + r)\} \subset \mathbb{R}^n$ to be analogous to the basic neighborhood system U_x we then have that $U \in \{B_r(x)\}$ is contained in $B_r(x)$ and contains x. We can reformulate our

definition of continuity at x in terms of this basic neighborhood system for X. We could say then that a function $f : X \to X$ is continuous at $x \in X$ iff $\forall V \in U_{f(x)}$, $\exists U \in U_x$ such that $f(U) \subset V$.

Definition *(Continuity on Basic Neighborhood Spaces)*

Let $(X, \mathcal{B}_X)$ and $(Y, \mathcal{B}_Y)$ be basic neighborhood spaces. If $\mathcal{B}_X = \{\mathcal{U}_x | x \in X\}, \mathcal{B}_Y = \{\mathcal{V}_y | y \in Y\}$ and $f : X \to Y$ then $f : (X, \mathcal{B}_X) \to (Y, \mathcal{B}_Y)$ is *continuous* at each $x \in X$ if for each basic neighborhood $V \in \mathcal{V}_{f(x)}$, there exists a basic neighborhood $U \in \mathcal{U}_x$ such that $f(U) \subset V$. If f is continuous $\forall x \in X$ then f is continuous on X. ♠

This is extremely useful as we have found a way to classify discrete spaces as continuous. This might seem a bit strange as if we took two points x and y in $\mathbb{R}$ we would have an infinite number of points between them. If however, for example, the universe is discrete then a better model would be the notion in which that of being infinitely divisible is replaced with two points being connected but having no points between them! Discrete continuity is important because as we can use the definition of *continuity* on these neighborhoods to relate morphisms of 'concepts' (**??**.

Example 0.5

Let $X = \{3, 7, 11\}$ *and let* $\mathcal{U}_3 = \{\{3\}, \{3, 7\}\}, \mathcal{U}_7 = \{\{7\}, \{7, 11\}\}, \mathcal{U}_{11} = \{\{7, 11\}\}$ *and let the function* $f : X \to X$ *be the function defined by* $f(3) = 3$, $f(7) = 7$, *and* $f(11) = 7$*; f is continuous at 3, 7, and 11, hence f is continuous on* X. ∎

Proof

(a) $f(3) = 3$. The two basic neighborhoods of $f(3) = 3$ are $\{3\}$ and $\{3, 7\}$. Since $\{3\} = U$ is a basic neighborhood of 3 with the property that $f(U) \subset \{3\}$ and $f(U) \subset \{3, 7\}$, f is continuous at 3.

(b) $f(7) = 7$. The two basic neighborhoods of $f(7) = 7$ are $\{7\}$ and $\{7, 11\}$ and as in part (a), since 7 is in both $\{7\}$ and $\{7, 11\}$, we have that $f(U)$, $U = \{7\}$ is a subset of both subsets of U_7.

(c) $f(11) = 7$. The only basic neighborhood of 11 is $\{7, 11\}$ and taking $f(\{7, 11\}) = \{f(7), f(11)\} = \{7\}$ which is a subset of a $V \in \mathcal{U}_i$ we have that f is continuous at 11.

Since we have that f is continuous on each of the points 3,7, and 11 $\in X$, f is continuous on X. ∎

Example 0.6 *Let X be a non-empty set that contains at least two elements, let $\mathcal{D}_X$ and $\mathcal{I}_X$ respectively denote the discrete and indiscrete basic neighborhoods on X. The function $f : (X, \mathcal{D}_X) \to (X, \mathcal{I}_X)$ defined by $f(x) = x \, \forall x \in X$ is continuous for all x hence is continuous on X. However the inverse mapping from the Indiscrete to the Discrete basic neighborhood spaces is not continuous.* ♦

Note Look at definition for open/closed based on Boundary Points (Wiki: Boundary-Topology)

Definition *(Open Set)*

A set $O \in X$ is called *open* if and only if O contains neighborhood of each of its points. ♠

Now, as a neighborhood $\mathcal{N}_p$ of a point p is defined to be a set containing an open subset of X which contains p, we have that an open set O must contain open sets containing each of its points.

Example 0.7 *Let $X = \{a, b, c\}$ and $\mathcal{T} = \{\emptyset, X, \{a\}, \{b\}, \{a, b\}, \{b, c\}\}$ The proper open sets in X are $\{\{a\}, \{b\}, \{a, b\}, \{b, c\}\}$* ♦

Definition *(Topology)*

Let X be a non-empty set, and let $\mathcal{T}$ be a non-empty family of subsets of X that satisfies the following properties:

$\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$;

if $O_\gamma \in \mathcal{T} \forall \gamma \in I$ an indexing set, then $\bigcup_{i \in \Lambda} O_\gamma \in \mathcal{T}$;

If $O_i \in \mathcal{T}, i = 1, 2, ..n$, then $\bigcap_i^n O_i \in \mathcal{T}$.

The family $\mathcal{T}$ is called a *topology* for the set X, and the ordered pair $(X, \mathcal{T})$ is called a *topological space* ♠

Definition *(Discrete Topology)*

Let X be a set. Let $\mathcal{D}_X$ be the set of unions and intersections of all possible subsets of X, including the null set $\{\emptyset\}$. We call $\mathcal{D}_X$ the *discrete topology*. ∎

Understanding of the discrete topology is extremely important! As we shall see, *concepts* .... are analogous to the kernel or cycle of a simplicial complex. This leads us to the idea of an indiscrete topology.

Definition *(Indiscrete Topology)*

Let X be some set. Let $\mathcal{I}_X$ be the set consisting of X and $\emptyset$. As all unions and intersections of elements of $\{\emptyset, X\}$ are also in $\mathcal{I}_X$ We have then that $\mathcal{I}_X$ is a Topology. We call this the *indiscrete topology* ∎

The significance of the indiscrete spaces plays an important role in the indistinguishability of points hence the inability for categorical hierarchy in the category of concepts.

Definition *(Continuous Topological Spaces)*

Let $(X, \mathcal{R})$ and $(Y, \mathcal{S})$ be two topological spaces and let $f : X \to Y$. We then have that $f : (X, \mathcal{R}) \to (Y, \mathcal{S})$ is *continuous* if and only if $\forall V \in \mathcal{S},\ f^{-1}(V) \in \mathcal{T}$. ♠

The notion of continuity is extremely important and is fundamental in Homology.

Definition *(Subspace (Relative/Induced) Topology)*

Let $S \subset X$. The *subspace topology* on S is defined by $\mathcal{T}_S = \{S \cap U | U \in \mathcal{T}\}$ and satisfies the axioms of a topology. The *subspace topology* is also called the *relative topology* or the *induced topology* ♠

We will see in chapter **??** the relation of the subspace topology to the original topology that we are able to distinguish asymmetric subsets of a topological built upon symmetric expressions. It is because of this ability to "distinguish" that the comparison of concepts is possible via propositional logic.

Definition *(Neighbourhood)*

If $(X, \mathcal{T})$ is a topological space and $p \in X$, then a *neighbourhood* of p is an open subset containing p.

Example 0.8 *Let* $X = \{3, 7, 11\}$ *and let* $\mathcal{T} = \{\emptyset, \{3\}, \{7\}, \{7, 11\}, \{3, 7\}, X\}$ *Let* $\mathcal{N}_3 = \{\{3\}, \{3, 7\}, \{3, 11\}, X\}$, $\mathcal{N}_7 = \{\{7\}, \{3, 7\}, \{7, 11\}, X\}$, *and* $\mathcal{N}_{11} = \{\{7, 11\}, X\}$. *In this example* $\{3, 11\}$ *is a neighbourhood of 3 but not of 11. Thus a set does not have to be a neighborhood of all of its points.* ♦

The above example is one of a set that is a neighourhood of one of its points, but not all of its points! This is important because we can have 'separation' between neighborhoods which will become more apparent comparing the differences between the conscious and the unconscious.

Definition *(Separated)*

Let $A, B \in \mathcal{T}$ a topological space. The sets A and B are said to be *separated* if there are neighborhoods X of A and Y of B such that $X \cap Y = \emptyset$. ♠

In any metric space two open balls $B_r(x_1)$ and $B_s(x_2)$ are separated whenever $d(x_1, x_2) \geq r + s$. Thus we have that any two separated open balls must be disjoint. This is important because in the case of the ultrametric property (see **??**), if two open balls (which are neighborhoods of each of their points) overlap, then they are points of the same open ball; hence not separated. This is, as will be seen, the underpinnings of Matte-Blanco's theory of the unconscious. Note that a T_1 space (separation of points) is not enough to distinguish points in an ultrametric space. In this space, to distinguish two points one needs the space to be Hausdorff or T_2 space.

It is not enough to have elements of a Topological space to be distinct (unequal), we would like them to be topologically distinguishable. Topological distinguishability among points is the case if points do not have the exact same neighborhood. Hence for two points to be distinguishable there must be an open set in which one point belongs to but the other does not. This is why, in the ultrametric topology, if two sets (open balls) have a point in common, then the neighborhoods become the same hence the points become indistinguishable.

Definition *(Cover)*

A *cover*C of a set X is a collection of sets whose union contains X as a subset. We call C an open cover if each of elements is an open set. ♠

Definition *(Presheaf on open sets)*

Let Op_X be the category of open sets (**??**) on X A *presheaf* on the open sets of X is a functor $F : Op_X^{op} \to Set$. ♠

Let $X = \bigcup U_i; U_i \in X$ a cover of X. A presheaf of sets is a functor F on X which associates to each open set $U \subset X$, a set $F(U)$, and to an open set $V \subset U$ a map $\rho_W : F(U) \to F(V)$ such that for each open inclusion $W \subset V \subset U$ one has $\rho_{UU} = id_U$. A presheaf over a space associates to each set an algebraic object. A presheaf of maps, sets, vector spaces, etc. is a set that evolves continuously over a topological space X.

0.3 Ultrametric

Motivation: A metric is a relationship between points to give the notions of distance. If we take two points in space, a space that we observe or that we just make up (after all even the space we observe is modeled by one that we have made) we would like to know things such as if the points are path connected. Ultimately we want to understand how thoughts and ideas are connected in our mental space as well as, for example, how close one idea is to another.

Definition *(Metric)*

A *metric* on a set X is a function:

$d : X \times X \to \mathbb{R}$ such that $\forall x, y, z \in X$

(1) $d(x, y) \geq 0$

(2) $d(x, y) = 0$ iff $x = y$

(3) $d(x, y) = d(y, x)$

(4) $d(x, y) \leq d(x, z) + d(z, y)$ ♠

Definition *(Metric Ball)*

Let (M, d) be a metric space; a set M with a metric d, a radius $r > 0$ centered at point $p \in M$ we define:

a *open metric ball*:

$B_r(p, r) = \{x \in M | d(x, p) < r\}$

and a *closed metric ball*:

$B_r[p,r] = \{x \in M | d(x,p) \leq r\}$ ♠

Metric balls allow us to take the idea of a distance metric and for our purposes make it n-dimensional. With balls we can give particular properties to each point in specific balls as well as mappings. This will be useful in categorizing objects such as ideas and their relation to other ideas.

Definition *(Ultrametric)* A metric d is called an *ultrametric* if for a set X and $\forall x, y, z \in X$

$$d(x,y) \leq max\{d(x,z), d(y,z)\}$$

♠

Now if the above is the case, then the length $d(x,y)$ can not be longer than $d(x,z)$ and $d(y,z)$. However we also need to look at the other possibilities i.e. $d(x,z) \leq max\{d(x,y), d(y,z)\}$ and $d(y,z) \leq max\{d(x,y), d(x,z)\}$ thus we could not have the case in which any particular distance metric $d(\cdot,\cdot)$ is greater than the other two. In the case of normed spaces considering the ultrametric property we can prove a stronger case using "Krull Sharpening":

Proposition 0.9 *$||x+y|| \leq max\{||x||, ||y||\}$ with equality if $||x|| \neq ||y||$.*

Proof *Wlog assume $||x|| > ||y||$. This implies that $||x+y|| \leq ||x||$. But we can also compute $||x|| = ||(x+y)-y||$ but since we have to satisfy this equality for all "sides" we also have that $||x|| \leq max\{||x+y||, ||y||\}$. We then have the equality $||x+y|| \leq ||x|| \leq max\{||x+y||, ||y||\}$. Now $max\{||x+y||, ||y||\} \neq ||y||$ as then $||x|| < ||y||$ which contradicts the assumption. Thus we have to have that $max\{||x+y||, ||y||\} = ||x+y||$. Therefore $||x+y|| \leq ||x|| \leq ||x+y||$ so we can see that $||x|| = ||x+y||$.* ∎

0.4 p-Adics

Motivation: p-adic numbers give us a different kind of metric which over a unique factorization domain (UFD) relates numbers via their unique product of primes. p-adic numbers are close when their difference is divisible by a high power of p - the higher the power they closer they are. Our most important use of the p-adic numbers is that this field of numbers satisfies ultrametric property. We can also create a topology on the p-adic open balls and this will be useful in our relationship to this topology and the indiscrete topology.

Definition *(p-Adic Valuation)*

A *p-adic valuation*, denoted $v_p(r)$ or $ord_p(r)$ is the largest value n such that $p^n|r$. If $r = \frac{a}{b}$ is rational then its p-adic valuation is defined as $v_p(r) = v_p(a) - v_p(b)$. ♠

Definition *(p-Adic Norm)*

For any non-zero number $x \in \mathbb{Q}$ there is a unique integer n allowing us to write $x = p^n(\frac{a}{b})$ where neither a or b is divisible by p. We define a *p-adic norm* of $x \neq 0$ to be $|x|_p = p^{-v_p(n)}$ and $|0|_p = 0$ if $x = 0$. ♠

Now by the fundamental theorem of arithmetic, for a given non-zero rational number x, there is a unique finite set of distinct primes $p_1, \cdots, p_r$ and corresponding non-zero integers $a_1, a_2, \cdots, a_r$ such that $x = \pm p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$. Thus we have that $|x|_{p_i} = p_i^{-a_i} \ \forall \ 1 \leq i \leq r$ and $|x|_p = 1 \, \forall \, p \notin \{p_1, p_2, ..., p_r\}$. In short, if a number n doesn't have p_i in its prime factorization then 0 is the greatest power and so $p_i^0 = 1$. Thus we can represent any rational number p-adically.

Example 0.10 *Let $x = 10, y = 24$, and $z = 25$. For this example we look at the 5-adic norm and have $|10|_5 = \frac{1}{5}, |24|_5 = 1$, and $|25|_5 = \frac{1}{25}$. We then can see that 10 and 25 are 5-adically closer than 24 and 25 as $\frac{1}{5} - \frac{1}{25} < 1 - \frac{1}{25}$.* ♦

Proposition 0.11 *p-adic numbers satisfy metric properties.*

Proof *If d denotes the a metric on the p-adic numbers and $d(m, o) = p^{-t_{mo}}$, $d(m, n) = p^{-t_{mn}}$ and $d(n, o) = p^{-t_{no}}$ We show that the metric axioms are satisfied:*

1. Let $d(x, y) = p^{-t_{xy}}$ since$t_{xy} \in \mathbb{R}$ the result follows.

2. Follows from the definition.

3. $d(x, y) = p^{-t_{xy}} = |x - y|_p = |y - x|_p = p^{-t_{yx}}$ which is the desired result.

4. For the triangle inequality we have: $x - z = (x - y) + (y - z)$ where $p^{t_{xy}}|(x - y)$ and $p^{t_{yz}}|(y - z)$. Then $x - z = p^{t_{xy}}k + p^{t_{yz}}j$. Without loss of generality assume $t_{xy} \geq t_{yz}$ so that

$$x - z = p^{t_{yz}}(p^{t_{xy} - t_{yz}}k + j)$$

If $t_{xy} > t_{yz}$, then $p^{t_{xy}-t_{yz}}k + j \equiv j \not\equiv 0 \pmod{p}$, so $t_{xz} = t_{yz}$. If $t_{xy} = t_{yz}$, however, then $x - z = p^{t_{yz}}(k + j)$, where $k + j$ may be divisible by p. In that case $t_{xz} > t_{yz}$. However, in all cases one finds that $t_{xz} \geq t_{yz}$ and hence that $d(x, z) \leq d(y, z)$. This shows that $d(x, z) \leq d(y, z) + d(x, y)$ ∎

Lemma 0.12 *$a, b \in \mathbb{Q}_l$*

1. $v_p(ab) = v_p(a) + v_p(b)$

2. $v_p(a + b) \geq min\{v_p(a), v_p(b)\}$

3. $v_p(a) \neq v_p(b) => v_p(a + b) = min\{v_p(a), v_p(b)\}$

Example 0.13 *We choose the numbers $p = 5$, $a = 8$, $b = 15$*
1. Let $v_5(120) = v_5(120) = n$ where $n = 1$ as 1 is the largest exponent of 5 in that $5^1|2^2 \times 3 \times 5$ Now $v_5(8) = 0$ since $5^0|2^3$ and $v_5(15) = 1$ as $5^1|3 \times 5$ hence $v_5(8) + v_5(15) = 0 + 1 = 1 = v_5(8 \times 15)$

2. $v_5(8 + 15) = v_5(23) = 0$ as 23 is a prime number so only $5^0|23$ Now $min\{v_5(8), v_5(15)\} = 0$ thus the equality holds.

3.Here $v_5(8) = 0 \neq v_5(15) = 1$ and as in 2. $v_5(8 + 15) = 0$ and $0 = \{v_5(8), v_5(15)\} = 0$. ♦

0.5 p-adic and the Ultrametric

Motivation: As we will eventually model the possibility that the unconscious is symmetric and will be using the ultrametric as a means to do so, we need a numbering system which satisfies the ultrametric property.

This next example shows that valuations satisfy the ultrametric inequality.

Example 0.14 *We need to calculate the ultrametric for $x_1, x_2, x_3 \in \mathbb{N}$ where p is a fixed prime in $\mathbb{N}$ with the values: $|x_1 - x_2|_p, |x_2 - x_3|_p$, and $|x_2 - x_3|_p$.*

Let $p = 3, x_1 = 5, x_2 = 8$, and $x_3 = 12$. We find that as $p|(|x_1 - x_2|)$ or that $3|(|5 - 8|)$ and that $-val_3(3) = -1$ likewise that $-val_3(|5 - 12|) = 0 = -val(|7-12|)$ thus we have that $3^{-1} \leq max\{3^0, 3^0\}$ and $3^0 \leq \{3^{-1}, 3^0\}$ which satisfies the ultrametric conditions. ♦

Proposition 0.15 *The set of p-adic numbers satisfy the ultrametric property.*

Proof By lemma 2.1 we have that $v_p(a+b) \geq min\{v_p(a), v_p(b)\}$. Wlog we said $min\{v_p(a), v_p(b)\} = v_p(a)$ hence $v_p(a+b) \geq v_p(a)$ therefore $-v_p(a) \geq -v_p(a+b)$. Now raising this to the $p'th$ power we have $p^{-v_p(a)} \geq p^{-v_p(a+b)}$ which means that $|a+b|_p \leq |a|_p$. ∎

It is important to note that for the p-adic metrics to follow the ultrametric relationships, the p-adic values must be related i.e. the differences between the combinations of the numbers $X, \neg X, x_3$ must be considered in the ultrametric; the points can not be any number!

Definition *(p-adic Ball $\mathbb{Q}_p$)* A *p-adic ball* is a metric ball which satisfies the p-adic norm. We define several different types of p-adic balls:

$$B^{=}(a,r) = \{x \in \mathbb{Q}_p; |x-a|_p = r\}$$

$$B^{\leq}(a,r) = \{x \in \mathbb{Q}_p; |x-a|_p \leq r\}$$

$$B^{<}(a,r) = \{x \in \mathbb{Q}_p; |x-a|_p < r\}$$

$$B^{c} = \{x \in \mathbb{Q}_p; |x-a|_p \geq r\}$$

$$B^{c\backslash=} = \{x \in \mathbb{Q}_p; |x-a|_p > r\}$$

♠

Proposition 0.16 *The Sphere $B^{=}(a,r)$ is an open set in $\mathbb{Q}_p$*

Proof Let $x \in B^{=}(a,r), \epsilon < r$. We will show that $B^{\leq}(x,\epsilon) \subset B^{=}(a,r)$. Let $y \in B^{\leq}(x,\epsilon)$. then $|x-y|_p < |x-a|_p = r$. As $|y-a|_p = |x-a|_p = r$ which means that $y \in B^{=}(a,r)$.

Proposition 0.17 *The open balls in $\mathbb{Q}_p$ are both open and closed.*

Proof Any open ball $B^{\leq}(a,r)$ is open in every metric space as any point $x \in B^{\leq}(a,r)$ is in $B^{\leq}(a,r)$ which is contained in $B^{\leq}(a,r)$. Now to prove that $B^{\leq}(a,r)$ is closed in $\mathbb{Q}_p$ we show that its compliment B^c is open. But $B^c = B^{c\backslash=} \cup B^{=}$. Now the set $B^{c\backslash=}$ is open in every metric space as if $y \in B^{c\backslash=}$ then $|y-a|_p = r_1 > r$. We claim that the open ball $B^{\leq}(y, r-r_1)$ is contained

in $B^{c\backslash=}$. If this were not the case then there would be an $x \in B^{\leq}(y, r_1 - r)$ such that $|x - a|_p \leq r$, however by the triangle inequality:

$$r_1 = |x - a|_p = |a - x + x - y|_p \leq |a - x|_p + |x - y|_p \leq r + r_1 - r = r_1$$

which is a contradiction. The proposition follows as the union of two open sets is open.

A point $x \in M$ is a boundary point of a set $A \subset M$ if any open ball centered at x

Proposition 0.18 *Two p-adic balls satisfying the ultrametric can not overlap without being the same.*

Proof Let $x \in B^{<}(a, r)$ and $|0|_p = 0$ and $\| \ \|$ be a non-Archimedian norm. We then have that $\| x - a \| < r$. As $x - b = (x - a) + (a + b)$ we have that $\| x - b \| \leq max(\| x - a \|, \| a - b \|) < r$ by the ultrametric property. We then have that $x \in B^{<}(b, r)$ and thus $B^{<}(a, r) \subseteq B^{<}(b, r)$ (see **??**). The proof in the other direction shows that $B^{<}(b, r) \subseteq B^{<}(a, r)$ hence if two open balls have a point in common then:

$$B^{<}(b, r) = B^{<}(a, r)$$

This is the same with closed balls $B^{=}(y, r)$ as well. ∎

0.6 Quantum Mechanics

Motivation: We want to model the Matte-Blanco's idea that consciousness is asymmetric. To do this we need to understand how one can create asymmetries, specifically those found in propositional logic. We will reach the point in which we can create these logical asymmetries, we however do so by the means of using quantum mechanical principles.

Definition *(Quantum System)*

A *quantum system* is a portion of the physical environment which is considered for quantum physics and having to do with the wave-particle duality. ♠

Definition *(Wavefunction)*

The *wavefunction* ψ of a given quantum system contains the measurable information about the system. ♠

Definition *(Quantum State (Eigenstate) and Vector (Eigenvector))*

A *quantum state* a superposition of a given Hermitian operator in which the squares of the coefficients are the probabilities that the system will be measured to be in a particular eigenstate. A quantum state is also called a *state vector* or *eigenvector*. ♠

A quantum state can either be 'pure' or 'mixed'.

Definition *(Pure Quantum State)*

A quantum system is in a *pure quantum state* if we have complete knowledge about the system. ♠

As a system is in a pure state if we have complete knowledge of the system and the state we are able to represent a pure state by a single ket-vector. Pure states can be interpreted in terms of probability densities in which one eigenvalue is one and the rest are zero. Thus we can say that a system is in a state $|\psi\rangle$ then $|\psi\rangle$ must be the only eigenstate of the probability density of the system ρ with eigenvalues equal to one. Mathematically we have to have $\rho|\psi\rangle = |\psi\rangle$. As any other state $|\phi\rangle$, orthogonal to $|\psi\rangle$, has a probability zero, hence it must be an eigenvector with eigenvalue zero, that is $\rho|\phi\rangle = 0$

Definition *(Mixed Quantum State)*

A *mixed quantum state* is a statistical mixture of more than one pure state. ♠

In other words a system is in a mixed state if there is only partial (or no) information of the system.

For mixed states, there is no unique measure of entanglement, entanglement always implies correlations between two systems, which must also exist in a coherent superposition. It is important to note though the presence of coherence nor correlations imply that a system is entangled.

Definition *(Observable or Eigenvalue)*

An *observable* is an eigenvalue of the the eigenstate (state vector) of the observable and corresponds to a physical quantity of a system. ♠

Example 0.19

$$Q|\psi\rangle = q_n|i\rangle \qquad (\star)$$

In the equation $(\star)$ The operator is denoted by Q. The wavefunction for a system contains the measurable information about the system. To obtain specific values (eigenvalues or observables) of the system, we operate on the wavefunction with the quantum mechanical operator Q which is associated with the parameter which we are looking to measure. These parameters are values such as energy, momentum, and as we will investigate, the 'truth values' associated with a 'logic wave' see **??**.

Definition *(Density Matrix)*

A *density matrix* ρ is a matrix which describes a system in a mixed state, that is; it describes a statistical mixture of several quantum states (also known as a quantum ensamble of states). ♠

We can write the probability density as a projection operator: $\rho = |\psi\rangle\langle\psi|$ as

$$\rho|\psi\rangle = (|\psi\rangle\langle\psi|)|\psi\rangle = |\psi\rangle(\langle\psi|\psi\rangle) = |\psi\rangle$$

Definition *(Hermitian)*

A square matrix is called *Hermitian* if it is self-adjoint, thus a Hermitian matrix A is one such that $A = A^\dagger$. ♠

Hermitian (self adjoint) operators play an important role in quantum mechanics; for example, an observable is always described by a self-adjoint operator. If A is a self-adjoint operator, then $\langle\psi|A|\psi\rangle$ is always a real number. This implies that expectation values of observables are real.

Definition *(Superimposed)*

Two states are *superimposed* it is possible for a system to take on either of the states. ♠

Example 0.20 *Starting with the basis* $|0\rangle$ *and* $|1\rangle$ *a system A can occupy both states i.e.* $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. *The possibilities for* $|0\rangle$ *and* $|1\rangle$ *are superimposed.* ♦

The superposition of $|X\rangle$ and $|Y\rangle$ means that there is the possibility of collapsing, upon measurement, to either of the states $|X\rangle$ or $|Y\rangle$. Before collapse this can be seen the weighted sum of both $|X\rangle$ and $|Y\rangle$. On the other hand when there is no superposition we have perfect distinguishability hence the definition.

Separability of states is possible when one is able to describe composite states as direct products of subsystem state vectors.

Definition *(Operator)*

An *operator* is a linear function that acts on the states of a system. ♠

0.7 Category Theory

Motivation: Matte Blanco's analysis of both the unconscious and conscious show similarities with category theory. More specifically he states many interesting properties concerning symmetries and objects of a class. Although category theory at that time was in its very beginning stages in the mathematical arena, and even now is only beginning to be explored in the areas of cognitive science and even further behind in psychology, it is our aim in this section to explore possible relationships between Blanco's observations and the basics of category theory as well as various applications.

Definition *(Category)*

A *category* $\mathcal{C}$ consists of:

1. A class $Ob(\mathcal{C})$ whose elements are called the *objects* of $\mathcal{C}$.

2. For each pair (A, B) of objects of $\mathcal{C}$ a set $Hom_{\mathcal{C}}(A, B)$ whose elements we call *morphisms* of A into B.

3. For each triple (A, B, C) of objects of $\mathcal{C}$ a map:

$$Hom_{\mathcal{C}}(B, C) \times Hom_{\mathcal{C}}(A, B) \to Hom_{\mathcal{C}}(A, C)$$

which is called *composition*. The axioms which a category must satisfy are:

1. If $A, B, C, D \in Ob(\mathcal{C})$ and $f : A \to B, g : B \to h : C \to D$ are morphisms, then $(hg)f = h(gf)$, that is, composition is associative.

2. For all $A \in Ob(\mathcal{C})$, there is an element of $Hom_{\mathcal{C}}(A, A)$, denoted as 1_A such that for every $B \in Ob(\mathcal{C})$, $f \in Hom_{\mathcal{C}}(A, B)$, and $g \in Hom_{\mathcal{C}}(B, A)$, the compositions $f \circ 1_A = f$ and $1_A \circ g = g$ hold.

♠

Example 0.21 *In the category of sets denoted* ***Set*** *the objects are sets. If* $A, B \in Ob(\boldsymbol{Set})$*, the morphisms* $f, g \in Hom_{\boldsymbol{Set}}$ *are all the morphisms* $f, g : A \to B$*. The composition is the composition of morphisms. Thus* ***Set*** *is a category in which* $1_A : A \to A$ *is the identity morphism defined by* $1_A(a) = a \, \forall a \in A$*.* ♦

Now as Matte-Blanco states: "Classes are differentiated means that there are asymmetric relations that differentiate them".

0.8 Simplicial Complexes

Motivation: The notion of a simplicial complex is a precursor to our attempts to build a 'Mental Space' in which are embedded 'Concepts'(see **??**). Concepts are modeled by simplicial complexes. In this section we look at what a simplicial complex (or just complex) is and how we can decompose larger simplicial complexes to smaller ones.

Definition *(Simplex)*

A *n-simplex* σ is a n-dimensional convex hull of $n + 1$ independent points $v_0, \ldots, v_n \in \mathbb{R}^n$

$$\sigma = \{\theta_0 v_0 + \cdots + \theta_n v_n | \theta_i \geq 0, 0 \leq i \leq n, \sum_{i=0}^{n} \theta_i = 1\}$$

♠

Definition *(Category of Simplical Complexes)*

The *category of simplicial complexes* denoted by Δ has objects ordered sets $[n] = \{0, 1, 2, \cdots, n\}$ and ordered preserving maps between the ordered sets. ♠

Definition *(Simplicial Complex and Sub-complex)*

A *simplicial complex* $\mathcal{K}$ is a finite collection of simplices such that for $\sigma \in \mathcal{K}$ and $\tau \subseteq \sigma$ implies that $\tau \in \mathcal{K}$, and $\sigma_0 \in \mathcal{K}$ implies $\tau \cap \sigma_0$ is either empty or a face of both. A *subcomplex* of $\mathcal{K}$ is a subset of $\mathcal{K}$ which is also a simplicial complex. ♠

Definition *(Face of a Simplicial Complex)*

The subcomplexes that belong to a complex $\mathcal{K}$ are called *faces* of the complex. A face Y is said to belong to a face X if $Y \subset X$. ♠

Example 0.22 *The tetrahedron* Υ *with the vertex set* $\{A, B, C, D\}$ *is a simplicial complex. Note that if we choose some simplicial set, say* $X = \{A, B, D\} \in \Upsilon$ *we have that* $\forall Y \subset X, Y \in \Upsilon$. ♦

Definition *(Simplical Map)*

Let Δ and Γ be abstract simplicial complexes. A *simplicial map* is a function f that maps the vertices of Δ to those of Γ and has the property that for any face X of Δ, $f(X)$, the image set, is a face of Γ. ♠

Definition *(Simplicial Set)*

A *simplicial set* X is a covariant functor:

$$X : \Delta^{op} \to \mathbf{Set}$$

that is, presheaves on Δ. ♠

Since we we define the category of concepts **??** to have concepts as objects constructions of other concepts and morphisms of inclusions in to other concepts we have a functor $F : \mathbf{Con} \to \mathbf{Set}$. We however have that the concepts are un-ordered hence that uniquness of concepts only depends on the unique construction of concepts via other concepts.

Definition *(Chain Complex)*

A *chain complex* $(C_\bullet, \partial_\bullet)$ for the boundary operator ∂ is a sequence of abelian groups or modules $\cdots C_2, C_1, C_0, C_1, C_{-2} \cdots$ which are connected by boundary operators $\partial_{i+1} : C_{i+1} \to C_i$ such that the composition of any two consecutive maps $\partial_i \circ \partial_{i+1} = 0$ for all i. ♠

Definition *(Boundary Homomorphism and Operator)*

We call ∂_k the *boundary operator* which maps the group of k-chains C_k on a simplicial complex $\mathcal{K}$ to C_{k-1}. That is:

$$\partial_k : C_k \to C_{k-1}$$

where ∂_k is the *boundary homomorphism*:

$$\partial_k(\sigma) = \sum_{i=0}^{k} (-1)^i (v_0, \cdots, \widehat{v_i}, \cdots, v_k)$$

on the oriented k-simplex $\sigma = (v_0, \cdots v_k), \sigma$ a basis element in C_k where $(v_0, \cdots, \widehat{v_i}, \cdots, v_k)$ is the i'th face of σ which is obtained by deleting its i'th vertex. ♠

Definition *(Boundary of a Chain Complex)*

Let C_{k+1} be the group of $(k+1)$-chains. The *boundary* written $B_k = im\, \partial_{k+1}$ is the boundary on C_{k+1}, that is $\partial_{k+1}(C_{k+1}) \subseteq C_k$. ♠

We will encounter some very interesting properties of boundaries on logical complexes.

Example 0.23 *Let X be a simplicial complex containing the chain $x = P_1P_2 + 2P_2P_3 + P_3P_1$. The boundary $B(X); \partial(X) = P_3 - P_2$ is a 0-boundary.* ♦

Definition *(Kernel and n-Cycle)*

The *kernel* ∂_n of of a chain complex C_n written $\partial_n(C_n)$ consists of those n-chains with boundary 0. The elements of the kernel are n-cycles, written as $Z_n(X)$ where the topological space X is a chain complex in C_n. ♠

Example 0.24 *Let $z = P_1P_2 + P_2P_3 + P_3P_1$ be an n-chain in a simplicial complex X. Applying the boundary homomorphism ∂_z we obtain*

$$\partial(P_1P_2 + P_2P_3 + P_3P_1) = P_2 - P_1 + P_3 - P_2 + P_3 - P_1 = 0$$

Hence $z \in Ker(\partial) = Z(X)$.

For another example $X \ni x = P_1P_2 + 2P_2P_3 + P_3P_1$, as seen above $\partial(X) = P_3 - P_2 \therefore x \notin Z(X)$. ♦

We will see that those n-chains with n-cycles; those chain complexes with a boundary 0 are extremely special. These particular chains will be those relating to the simplicial complexes which are "concepts".

1 Mathematical-Logic Constructions

Motivation: In advancing upon our goal of relating propositional logic to quantum mechanics we must be able to find a correspondence between the two. This does not only entail how they relate, (as we shall see operators can be viewed as the superposition of other operators), but to be able to move between the two fields mathematically.

In this section we will proceed to develop the necessary mathematical constructions in order to relate logic to quantum mechanics. We start by partitioning sets of logical expressions such that we can rewrite all logical expressions as a combination of disjoint logical expressions. We then create vectors from these logical expressions and then move on to being able represent logical expressions in a linear form in a way analogous to vectors in quantum mechanics. We do this by the development of logical 'correspondence matrices' and 'bra-kets'. We then move on to more abstract structure such as tensors of logical expressions. Lastly, in **??** we show how these logical expressions act if similar

1.1 Logic Representations

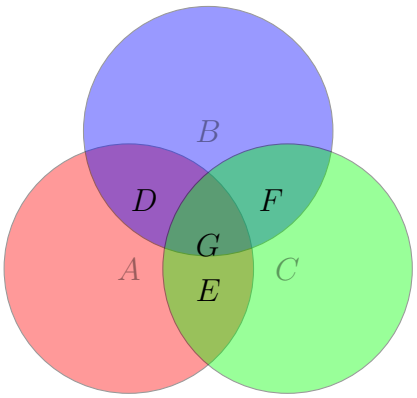
1.1.1 Partitioned Basis

Motivation: When dealing with the OR operator in propositional logic there are many possible overlaps between sets, this can make it difficult to distinguish particular elements without writing a long logical expression which will allow us to partition the sections of our potentially overlapping sets. By choosing a basis, one which is a partitioning of the universal set, we rid ourselves of conjunctions between partitions! This is extremely powerful and as we will see in the sections on topology (and thus a lot of the paper which makes use of topological spaces), we can easily create a topology as well as simplify the expressions of pre-sheaves and sheaves among many other things throughout the paper.

Our goal in this section is to obtain a way to remove the possible overlaps between sets; that is, we want to find a set (a basis) of logical expressions which allow us to remove the OR operator. Consider the following basis of

a Venn diagram over 3 logical variables X, Y, and Z.

A.	$X \wedge \neg(Y \vee Z)$
B.	$Y \wedge \neg(X \vee Z)$
C.	$Z \wedge \neg(X \vee Y)$
D.	$(X \wedge Y) \wedge \neg Z$
E.	$(X \wedge Z) \wedge \neg Y$
F.	$(Y \wedge Z) \wedge \neg X$
G.	$X \wedge Y \wedge Z$
H.	$\neg X \wedge \neg Y \wedge \neg Z$



It can be seen that if one were to combine (via OR) the first seven of the expressions (i.e. $A, \cdots, G$) one ends up with $X \vee Y \vee Z$ and if one also OR's the 8th expression H to the other seven, one has a Tautology. We denote $\mathcal{S} = \{A, B, .., H\}$, but we also overload our notation and let the comma separated values be the representation for the disjunction of $A, B, ..H$ e.g. $A \vee B \vee .. \vee H$. Hence all of the expressions in $\mathcal{S}$ are OR'ed together.

One can check that the each of the expressions $A, \cdots, H$ have exactly one truth valuation resulting in true, but furthermore that each of these valuations in which the result is true is distinct from every other, that is, a single row in the truth table has a valuation resulting in truth for a particular truth valuation on the set of logical variables. The set of these particular valuations is a partition.

The reason that we chose to represent the logical expressions as a partition is that we then are given the ability to look at disjoint expressions (we will see that these are sections of a sheaf) whereas in the case of just 3 variables X, Y, Z there is both the possibility of overlap between the variables which make this formulation of logical expressions more difficult to represent for what we would like to show.

It is important to note: that, in general, there are $2^k = n$ sections of a k-variable Venn diagram and 2^{2^k} possible subsets of sections which correspond

to $\sum_{i=0}^{k} \binom{2^k}{i}$ possible combinations of choosing combinations of partitioned sections.

1.1.2 Logical Vectors

Motivation: This section develops the idea of 'Logical Vectors'. These are structures are developed for the purpose of relating logical expressions to vectors as one would find in use in quantum mechanics.

In this section we define the basic notions such as 'logical points', 'logical vectors', and a 'logical vector space' in which to work.

Definition *(Boolean Matrix)*

A *boolean matrix* is a matrix in which all entries are either 0 or 1. ♠

Definition *(Logic Vector)*

A *logic vector* is an ordered collection of n elements, each element of which is an element of a boolean algebra. ♠

Definition *(Logic Vector Space)* A *logic vector space* is a set of logic vectors which follow the rules of a vector space. ♠

Definition *(Logical Basis Element)*

Let X be a logical variable. We call X_T and X_F *logical basis elements* of X where X_T denotes when $\mathcal{V}(X) = T$ and X_F denotes when $\mathcal{V}(X) = F$. ♠

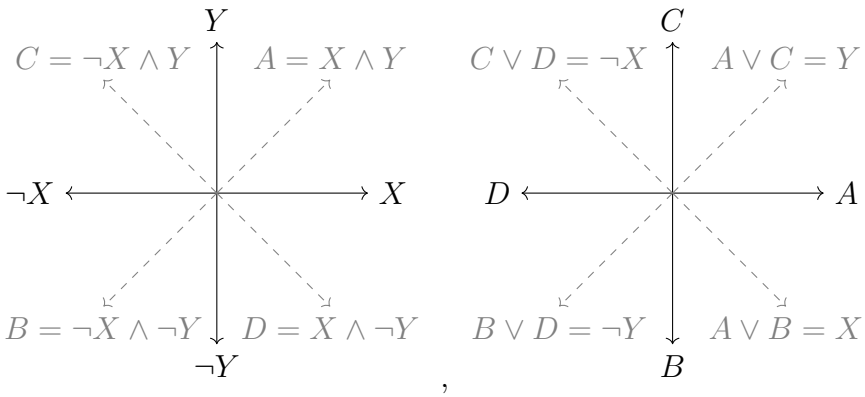
Definition *(Logic Point)*

A *logic point* P consists of an ordered set of basis elements. We will denote a logic point in the form $P = (\mathcal{V}(X_1), \mathcal{V}(X_2), \cdots, \mathcal{V}(X_n))$ where $\mathcal{V}(X_i)$ is the truth valuation on X_i. ♠

Example 1.1 *Letting* $A = X \wedge Y$, $B = X \wedge \neg Y$, $C = Y \wedge \neg X$, *and* $D = \neg(X \vee Y)$ *we see the following truth table for several logical vectors:*

A	B	C	D	*Logical Vector*
T	T	F	F	$A \vee B = X$
T	F	T	F	$A \vee C = Y$
F	T	T	F	$B \vee C = X \Updownarrow Y$
T	F	F	T	$A \vee D = X \Leftrightarrow Y$

That is, each of the literals $L \in \{A, B, C, D\}$ which has a valuation equal to true i.e. $\mathcal{V}(L) = T$ gives this element in the conjunction. Letting the tuple of We have a set of 4 points $\{(T, T, F, F), (T, F, T, F), (F, T, T, F), (T, F, F, T)\}$



of the form $(\mathcal{V}(A), \mathcal{V}(B), \mathcal{V}(C), \mathcal{V}(D))$ for the vector $(A, B, C, D) = A \vee B \vee C \vee D$. ⧫

We choose to represent the logical expressions over variables such as X, Y in this way as by doing so we are able to distinguish each point from the others. We are able to do this due to the basis, such as A, B, C, D in the above example due to the fact that logical AND is unused. This can be seen as all our newly chosen basis elements are such that $R \wedge S = F,\ R, S \in \{A, B, C, D\}$. If however we took the $2^2 = 4$ points in just two basis elements X, Y using only $\vee$ we would not be able to represent the conjunction of these elements, to do so we would need $\wedge$ hence a possibility of 4 other points. The combination of all points over both $\vee$ and $\wedge$ give $4 \cdot 4 = 16$ possibilities for logical expressions. This is precisely what we have in our basis of 4 elements as each can either be true or false, hence $2^4 = 16$ possibilities. We will make extensive use of this new basis for our logical vectors especially in building topological spaces as well as the accompanying homological algebra between them. We will also employ these base transformations in the Quantum analogs.

1.1.3 Correspondence Matrices and Bra-Kets

Motivation: Quantum Mechanics (QM) employs its own notation (i.e. Bra-Ket's) in its treatment of ideas which we build on in this thesis. In this section we encode logical expressions in the notation used in QM. This is partly to manipulate expressions as is done in QM, but also as we will discover; we can treat the logical operator correspondence matrices similar to operators in QM. But also note the interesting ability to transform one logical expression in to another via matrix rotation!

Correspondence matrices give us a way to represent operators in propositional logic using matrices. This ability will be indispensable as by representing the logical operators by matrices (which are linear) not only will we be able to speed up computations on logical expressions we will be able to build a foundation for what we will call logical simplices and continue to build a "mental space".

In this section we develop a set of 16 representations of operators in propositional logic. We also develop a linear representation 'Bra-Ket' which is native to quantum mechanics.

First, we choose a suitable basis to represent our logical variable. Since we are only concerned in whether the variable is true or false, we choose a basis whose two basis elements are opposite one another. That is, for all logical variables X, $X = span\{X, \neg X\}$ where $X = \neg\neg X$. We proceed to encode the information by the use of linear vector spaces using the bra-ket notation. This notation is chosen specifically as it is particular to quantum mechanics which we will make heavy use of later in this thesis.

Definition *(Logical Bra and Ket Vectors)*

Let B be a Boolean vector space and X a logical vector in B. The row vector $\langle X| = [X, \neg X]$ is called a $bra - vector$ and column vector $|X\rangle = [X, \neg X]^T$ is called a $ket - vector$. ♠

An example of the bra-ket notation is given in the following definition:

Definition *(True and False Vectors)*

For $X = 1$ and $X = 0$ where $\neg 1 = 0$ we define the *true ket-vector* and *false ket-vector* respectively as follows:

$$|1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } |0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

♠

As one can see, if $|X\rangle = |1\rangle = \begin{bmatrix} 1\,0 \end{bmatrix}^T$ then $X = 1$ and $\neg X = 0$ and as $\neg 1 = 0$ and $\neg 0 = 1$ the false ket-vector $|0\rangle = |X\rangle \Rightarrow X = 0$ and $\neg X = 1$ as we would expect.

Definition *(Correspondence Matrix)*

Let X and Y be logical literals. The *correspondence matrix* is the matrix derived from the tensor product of base vectors. Let

$$X\rangle = \begin{bmatrix} X \\ \neg X \end{bmatrix} \text{ and } |Y\rangle = \begin{bmatrix} Y \\ \neg Y \end{bmatrix}$$

The *correspondence matrix* is

$$|X\rangle \otimes |Y\rangle = \begin{bmatrix} XY & X\neg Y \\ \neg XY & \neg X\neg Y \end{bmatrix}$$

in which for each position in the matrix $X_i Y_j = X_i \wedge Y_j$ are elements in the truth table $\mathbb{T}$. ♠

The following are the four outer (tensor) products of truth and false ket vectors:

$$|1\rangle \otimes |1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, |1\rangle \otimes |0\rangle = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, |0\rangle \otimes |1\rangle = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, |0\rangle \otimes |0\rangle = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Definition *(AND [$\wedge$] Correspondence Matrix)*

We define the *AND correspondence matrix* written as [$\wedge$] to be

$$|1\rangle \otimes |1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Example 1.2 *To see how we find the corresponding logical expression from* $[\wedge]$ *we multiply this matrix by vectors* $|X\rangle^T$ $|Y\rangle$*:*

$$[X, \neg X] \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} Y \\ \neg Y \end{bmatrix} = XY \rightsquigarrow X \wedge Y$$

Which in view of the relation defined above $\cdot \rightarrow \wedge$ *we have our logical expression* $X \wedge Y$*.*

Definition *(OR* $[\vee]$ *Correspondence Matrix)*

$$|1\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle + |0\rangle \otimes |1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

X	Y	$X \vee Y$
0	0	0
0	1	1
1	0	1
1	1	1

Definition *(Right Implication* $[\Rightarrow]$*) Correspondence Matrix*

The *right implication correspondence matrix*, written $[\Rightarrow]$ corresponds to the matrix: $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. We write the operation of $[\Rightarrow]$ on vectors X and Y as $\langle X|[\Rightarrow]|Y\rangle$. ♠

Example 1.3 *Working out the computation of right implication on two vectors* $X = [X, \neg X]$ *and* $|Y\rangle = [Y, \neg Y]$

$$: \langle X|[\Rightarrow]|Y\rangle = [X, \neg X][\Rightarrow][Y, \neg Y]^T = -X + Y$$

Which then has the logic analog of $\neg X \vee Y$*. The truth table is:*

$\neg X$	Y	$-X \vee Y$
0	*0*	*0*
0	*1*	*1*
1	*0*	*1*
1	*1*	*1*

We can see that the only time the dis-junction is 0 is when $\neg X = 0\,(X = 1)$ *and* $Y = 0$. *Which means that* $|X\rangle$ *= 1 and* $|Y\rangle$ *= 0.* ♦

It is important to note that we can not just pull out the negative sign in the matrix.

Definition *(Left Implication* $[\Leftarrow]$ *Correspondence Matrix)*

The *left implication correspondence matrix*, written $[\Leftarrow]$ corresponds to the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. We write the operation of $[\Leftarrow]$ on two vectors X and Y as $\langle X|[\Leftarrow]|Y\rangle$. ♠

Definition *(Up Implication* $[\Uparrow]$ *Correspondence Matrix)*

The *up implication correspondence matrix*, written $[\Uparrow]$ corresponds to the matrix: $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. We write the operation of $[\Uparrow]$ on two vectors X and Y as $\langle X|[\Uparrow]|Y\rangle$. ♠

Definition *(Down Implication* $[\Downarrow]$ *Correspondence Matrix)*

The *down implication correspondence matrix*, written $[\Downarrow]$ corresponds to the matrix: $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$. We write the operation of $[\Downarrow]$ on two vectors X and Y as $\langle X|[\Downarrow]|Y\rangle$. ♠

Definition *(Symmetric Implication* $[\Leftrightarrow]$ *Correspondence Matrix)*

The *symmetric implication correspondence matrix* (*SImp*), written $[\Leftrightarrow]$ corresponds to the matrix: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. We write the operation of $[\Leftrightarrow]$ on two vectors X and Y as $\langle X|[\Leftrightarrow]|Y\rangle$.♠

Definition *(Asymmetric Implication* $[\Updownarrow]$ *Correspondence Matrix)*

The *asymmetric implication correspondence matrix* (*ASImp*), written $[\Updownarrow]$ corresponds to the matrix: $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. We write the operation of $[\Updownarrow]$ on two vectors X and Y as $\langle X|[\Updownarrow]|Y\rangle$.♠

Note that ASImp corresponds to the symmetric difference $\oplus$, the reason different notation ($\Updownarrow$) is used is that it better reflects the change in symmetry via matrix rotation.

Definition *(Impax $[\Updownarrow]$ Correspondence Matrix)*

The *impax implication correspondence matrix*, written $[\Updownarrow]$ corresponds to the matrix: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. We write the operation of $[\Updownarrow]$ on two vectors X and Y as $\langle X|[\Updownarrow]|Y\rangle$.♠

It is important to note that $[\Updownarrow] = [\Leftrightarrow] + [\Updownarrow]$

Op.	Matrix	Op.	Matrix	Op.	Matrix	Op.	Matrix
$[\Updownarrow]$	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	$[0]$	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	$[\Leftrightarrow]$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$[\Updownarrow]$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
$[\wedge]$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$[\neg\vee]$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	$[\Uparrow]$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$[\Downarrow]$	$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$
$[\Rightarrow]$	$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$	$[\Leftarrow]$	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$[\vee]$	$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$	$[\neg\wedge]$	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
$[R]$	$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	$[\neg R]$	$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$	$[L]$	$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$	$[\neg L]$	$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$

Figure 2: The set of 16 distinct correspondence matrices

Example 1.4 *Let $X \vee Y$ and $X \wedge Y$ be two logical expressions. Consider the expression $(X \vee Y) \setminus (X \wedge Y)$. Computation shows that this is equivalent to $(\neg X \vee \neg Y) \wedge (X \vee Y) = \langle X|[\Updownarrow]|Y\rangle$.*

Regarding the corresponding matrix representations we see that:

$$\langle X|[\vee]|Y\rangle \circledwedge \langle X|[\neg\wedge]|Y\rangle = \langle X|[\Updownarrow]|Y\rangle$$

♦

It is interesting to note that $(X \wedge Y) \setminus (X \vee Y) = \langle X|[\neg\wedge]|Y\rangle \circledwedge \langle X|[\vee]|Y\rangle = [0]$

In this last example we are taking $(X \vee Y) \setminus (X \wedge Y) = (X \vee Y) \wedge \neg(X \wedge Y)$.

1.1.4 Tensors and Bra-Kets

Motivation: Tensors are used in many areas of mathematics such as quantum mechanics and general relativity. Many times because the amount of information becomes difficult to write out in large matrices, but very often one needs to transform from one set of basis vectors to another.

In this section we develop a way to represent logical expressions (vectors) as tensors as well as show how they are used. Furthermore we continue the development of the bra-ket notation and show many interesting properties which they satisfy. Note the correspondences at the end of the chapter.

We use subscripts and superscripts to denote the row elements and column elements respectively. We then let $\langle X| = X_i = [X_1, X_2]$ where $X_1 = X, X_2 = \neg X$ and $|Y\rangle = Y^j = [Y^1, Y^2]^T$ where $Y_1 = Y, Y_2 = \neg Y$. If we want to represent the outer product matrix $|Y\rangle\langle X|$ in tensor notation we write the outer product matrix $Y^j \otimes X_i = Z_i^j$. Thus we can see that for an M_l^k matrix we can represent the row and column vectors by the lower and upper indices respectively. Since none of our matrices are complex we do not need to worry about the complex conjugation as well as the transpose.

Example 1.5 *Consider the expression* $\langle X|[A]|Y\rangle\langle Y|[B]|Z\rangle$

$$|Y\rangle\langle Y| = |Y\rangle \otimes \langle Y| = \begin{bmatrix} Y^1 \\ Y^2 \end{bmatrix} \otimes \begin{bmatrix} Y_1 & Y_2 \end{bmatrix} = \begin{bmatrix} Y^1Y_1 & Y^1Y_2 \\ Y^2Y_1 & Y^2Y_2 \end{bmatrix}$$

As $Y \neq \neg Y$ *and we just have basis elements of 0 and 1, this would leave either* Y *or* $\neg Y$ *(not both) to be 0. Thus we have that* $Y^1 \otimes Y_2 = Y^2 \otimes Y_1 = 0$. *Furthermore as* $Y^i \otimes Y_i = Y_i^i$ *and* $d(Y)$ *is the diagonal operator on the matrix* Y. *Let:*

$$A = \begin{bmatrix} A_1^1 & A_1^2 \\ A_2^1 & A_2^2 \end{bmatrix},\ B = \begin{bmatrix} B_1^1 & B_1^2 \\ B_2^1 & B_2^2 \end{bmatrix} \text{ and } [d(Y)] = \begin{bmatrix} Y_1^1 & 0 \\ 0 & Y_2^2 \end{bmatrix}$$

We then have the expression $\langle X|[A][d(Y)][B]|Z\rangle = X_i A_j^i Y_k^j B_l^k Z^l$ *and as* $[d(Y)]$ *is diagonal* $X_i A_j^i Y_k^j B_l^k Z^l \delta_k^j = X_i A_j^i Y_j^j B_l^j Z^l$. ♦

Example 1.6 *Let* $[A] = [\Rightarrow] = [B]$, *we then have the expression:*

$$\langle X|[\Rightarrow]|Y\rangle\langle Y|[\Rightarrow]|Z\rangle = X_i [\Rightarrow]_j^i Y_j^j [\Rightarrow]_l^j Z^l$$

Which we can look at component-wise in the table:

i	j	l	$X_i A_j^i Y_j^j B_l^j Z^l$	$X_i Y_j^j Z^l$	$X_i \wedge Y_j \wedge Z_l$
1	*1*	*1*	$X_1 \cdot 1 \cdot Y_1^1 \cdot 1 \cdot Z^1$	$X_1 Y_1^1 Z^1$	$X \wedge Y \wedge Z_1$
1	*1*	*2*	$X_1 \cdot 1 \cdot Y_1^1 \cdot 0 \cdot Z^2$	*0*	*0*
1	*2*	*1*	$X_1 \cdot 0 \cdot Y_2^2 \cdot 1 \cdot Z^1$	*0*	*0*
1	*2*	*2*	$X_1 \cdot 0 \cdot Y_2^2 \cdot 1 \cdot Z^2$	*0*	*0*
2	*1*	*1*	$X_2 \cdot 1 \cdot Y_1^1 \cdot 1 \cdot Z^1$	$X_2 Y_1^1 Z^1$	$\neg X \wedge Y \wedge Z_1$
2	*1*	*2*	$X_2 \cdot 1 \cdot Y_1^1 \cdot 0 \cdot Z^2$	*0*	*0*
2	*2*	*1*	$X_2 \cdot 1 \cdot Y_2^2 \cdot 1 \cdot Z^1$	$X_2 Y_2^2 Z^1$	$\neg X \wedge \neg Y \wedge Z_1$
2	*2*	*2*	$X_2 \cdot 1 \cdot Y_2^2 \cdot 1 \cdot Z^2$	$X_2 Y_2^2 Z^2$	$\neg X \wedge \neg Y \wedge Z_2$

One can check that:

$(X \wedge Y \wedge Z) \vee (\neg X \wedge Y \wedge Z) \vee (\neg X \wedge \neg Y \wedge Z) \vee (\neg X \wedge \neg Y \wedge \neg Z)$

$= (Y \wedge Z) \vee (\neg X \wedge \neg Y) = (\neg X \vee Y) \wedge (\neg Y \vee Z) = (X \Rightarrow Y) \wedge (Y \Rightarrow Z)$. ♦

Looking deeper in to the expression $X_i A_j^i Y_j^j B_l^j Z^l$ we can see that it is equivalent to $(X_i A_j^i Y_j)(Y^j B_l^j Z^l)$. This hints at the ability to break apart the matrix multiplication which, as multiplication has the analog of AND'ing together two expressions we can equivalently AND together two matrices if we give an 'outer AND' (analogous to the outer product), which we call the 'direct AND'.

Definition (Direct AND)

The direct AND of two representations ρ_1 and ρ_2 written $\barwedge$ is the component wise AND:

$$\rho_{1_{ij}} \barwedge \rho_{2_{ij}} = \begin{bmatrix} \rho_{1_{11}} \wedge \rho_{2_{11}} & \rho_{1_{12}} \wedge \rho_{2_{12}} \\ \rho_{1_{21}} \wedge \rho_{2_{21}} & \rho_{1_{22}} \wedge \rho_{2_{22}} \end{bmatrix} \text{ for } i, j \in \{1, 2\}$$

♠

Definition (Direct OR)

The direct OR of two representations ρ_1 and ρ_2 written $\barvee$ is the component wise OR:

$$\rho_{1_{ij}} \barvee \rho_{2_{ij}} = \begin{bmatrix} \rho_{1_{11}} \vee \rho_{2_{11}} & \rho_{1_{12}} \vee \rho_{2_{12}} \\ \rho_{1_{21}} \vee \rho_{2_{21}} & \rho_{1_{22}} \vee \rho_{2_{22}} \end{bmatrix} \text{ for } i, j \in \{1, 2\}$$

♠

The next example involves an equation which is of fundamental importance in the upcoming model in which we will see how breaking down symmetry can give asymmetry. The following example shows how transposition can be applied to the logical correspondences but manipulated through tensors.

Example 1.7 *Consider the expression:* $\langle X|[\Rightarrow]|Y\rangle\langle X|[\Leftarrow]|Y\rangle$

We let $(\langle X|[\Leftarrow]|Y\rangle)^T = \langle Y|[\Rightarrow]|X\rangle$. *This can be seen as* $|Y\rangle^T = \langle Y|$, $\langle X|^T = |X\rangle$ *and* $[\Leftarrow]^T = [\Rightarrow]$. *We then have the equivalent expression:*

$$\langle X|[\Rightarrow]|Y\rangle\langle Y|[\Rightarrow]|X\rangle = \langle X|[\Rightarrow]|Y\rangle\langle Y|[\Rightarrow]|X\rangle = \langle X|[\Leftrightarrow]|Y\rangle$$
$$= (X \Rightarrow Y) \wedge (Y \Rightarrow X) = X \Leftrightarrow Y$$

We can also see this by examining the inner matrices first, that is;

$$[\Rightarrow]|[Y]\rangle\langle[Y]|[\Rightarrow] = [\Rightarrow]\textcircled{\wedge}[d(Y)]\textcircled{\wedge}[\Rightarrow] = [d(Y)]$$

where $[d(Y)]$ *is the 2 by 2 diagonal matrix for the logical variable* Y. *We then compute:*

$$\langle X|[[d(Y)]]|X\rangle = X \Leftrightarrow Y$$

Another extremely useful way of showing this is by considering the commutative of $\textcircled{\wedge}$. *Taking the original expression without transpose and we find:*

$$\langle X|[\Rightarrow]|Y\rangle\langle X|[\Leftarrow]|Y\rangle = \langle X|Y\rangle\langle X|[\Rightarrow][\Leftarrow]|Y\rangle$$

As $\langle X|Y\rangle = \langle X|[\Leftrightarrow]|Y\rangle$ *and* $[\Rightarrow]\textcircled{\wedge}[\Leftarrow] = [\Leftrightarrow]$ *that we have the original expression equivalent to:*

$$\langle X|[\Leftrightarrow]|Y\rangle\langle X|[\Leftrightarrow]|Y\rangle = \langle X|[\Leftrightarrow]|Y\rangle$$

This is of course because elements of a boolean algebra are idempotent. ♦

We can see that $[\Rightarrow]^T \neq [\Rightarrow]$ however $[\Leftrightarrow]^T = [\Leftrightarrow]$. This is extremely important as it will allow us to classify expressions in respect to their asymmetries and symmetries. The following logical transformations, one of which which can be seen in the last example, allow us to transform representations of logical expressions in to equivalent representations. This allows for the factorizing of linear vectors so as to simplify the representation of logical expressions.

In the following we will denote $[\Phi]^{n^\circ}$ as the clockwise rotation of the logical matrix $[\Phi]$. We have the relations:

$$\langle X|[\Phi]|\neg Y\rangle = \langle X|[\Phi]^{90^\circ}|Y\rangle$$
$$\langle \neg X|[\Phi]|\neg Y\rangle = \langle X|[\Phi]^{180^\circ}|Y\rangle$$
$$\langle \neg X|[\Phi]|Y\rangle = \langle X|[\Phi]^{270^\circ}|Y\rangle$$
$$\neg \langle X|[\Phi]|Y\rangle \ \langle Y|[\neg\Phi]^T|X\rangle$$

These correspondences will be useful changing a logical expression from it's negated state to that of the non-negated state and a new logical operator.

1.2 Quantum like Logic

Motivation: As we look to understand the unconscious vs. the conscious in the mind we look to the symmetries vs. asymmetries respectively. We would like to see how one may go from the symmetric state to that of the asymmetric state in logic. To do this we employ a methods similar to those in quantum mechanics especially that of superposition, but applied to that of logical operators.

In this section we will develop quantum mechanical ideas such as 'measurement' and 'superposition' in terms of logical operators on 'logical waves'.

We start by defining some quantum like logical correspondences.

Definition *(Quantum-Logical AND)*

We define the $Quantum - Logical\ AND$ of $X \wedge Y$ to be $|X\rangle \circledwedge |Y\rangle$ ♠

Definition *(Quantum-Logical OR)*

We define the $Quantum - Logical\ OR$ of $X \vee Y$ to be $|X\rangle \circledvee |Y\rangle$ ♠

Definition *(Projection Operator)*

Given two logic vectors α and β, we define the *projection operator* $[P]$ to be $[P] = |\alpha\rangle\langle\beta|$ ♠

Definition *(Projection)*

Let $|\psi\rangle$ be a logic vector and $[P]$ a projection operator. The *projection* of $|\psi\rangle$ along $[P]$ is written $[P]|\psi\rangle$ ♠

Definition *(Measurement)*

We call a *measurement*, or more formally, *projective measurement* the result of the action of a projective operator on a logic vector i.e. a projection. ♠

Definition *(Logic Wavefunction [Logic Wave])*

A *logical wave* or *logic wave* or just *L-wave* for short, is the superposition of true and false valuations. In notation where L is a logical expression:

$$|\mathcal{V}(L)\rangle = |\mathcal{V}_T(L)\rangle \circledcirc |\mathcal{V}_F(L)\rangle$$

Definition *(Logical Superposition)*

We define the *logica superposition* of two logical expressions $L_1, L_2 \in \mathbb{L}$ to be $|L_1\rangle \circledcirc |L_2\rangle$. ♠

Definition *(Quantum-Logical Negation)*

We define the $Quantum - Logical\ Negation$ of $\neg X$ to be $\neg|X\rangle{=}|\neg X\rangle$ ♠

Example 1.8

Using the basis $|1\rangle = [1,0]^T$ *and* $|0\rangle = [0,1]^T$. *We calculate* $|0\rangle \otimes |1\rangle = [0,1]^T \otimes [1,0]^T = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ *But this is equal to* $(|0\rangle \otimes |1\rangle)^T$ *which gives us* $\langle 0| \otimes \langle 1|$. *We can see that* $\langle 1|1\rangle = \langle 0|0\rangle = 1$ *and* $\langle 1|0\rangle = \langle 0|1\rangle = 0$. ♦

We can view $|X\rangle = X|1\rangle + \neg X\ |0\rangle$ as we can see that $X|1\rangle = [X,0)^T$ and $\neg X\ |0\rangle = (0,\neg X]^T$. Thus we have that a logical atom $\phi \in \mathcal{L}$ can be decomposed in to perpendicular basis ϕ_1 or ϕ_2 hence $\phi = \phi_1 \vee \phi_2$. This is a particularly insightful example as if we don't know the the valuation of ϕ we don't know whether $\mathcal{V}(\phi_1) = T$ or $\mathcal{V}(\phi_2) = T$. We could say that in a sense,

if looked at as an overlapping of both possibilities of ϕ_1 and ϕ_2 i.e. $\phi_1 \vee \phi_2$.

Let us examine the expression $\mathcal{V}(X \vee Y) = T$. This means that we have any of the cases; that of which $\mathcal{V}(X) = T$, $\mathcal{V}(Y) = T$, which of course implies that $\mathcal{V}(X \wedge Y) = T$ if the truth valuation of X and Y is true. Thus if $\mathcal{V}(X \vee Y) = T$ we don't know, without being able to distinguish between both X and Y. In other words, if we obtain a value of true from the conjunction of two logical variables, we can not distinguish the individual values of the logical constituents without some other measurement. We will call two logical expressions which are conjoined to be logically indistinguishable.

Definition *(Logically Indistinguishable [Over Truth])*

Let $L_1, L_2 \in \mathcal{L}$ a set of expressions in propositional logic. We call L_1 *logically indistinguishable* over truth from L_2 if $\mathcal{V}(L_1 \vee L_2) = T$ ♠

Note that this works also for $L_1, L_2 \in \mathcal{L}$ in which $L_2 = \neg L_1$. Indistinguishably among expressions in propositional logic raises the question of when two expressions are distinguishable. Intuitively one may see that two logical expressions are distinguishable (over a truth value) when their truth values corresponding to the satisfiability for a valuation giving a particular truth value can be distinguished. In other words, for $\mathcal{V}(L_1 \wedge L_2) = T$ to be satisfiable we must have that $L_1 = L_2 = T$. Thus we know what the precise the values of L_1 and L_2 must be; that there is no ambiguity among possible truth assignments leading to a given valuation.

Definition *(Logically Distinguishable [Over Truth])*

Let $L_1, L_2 \in \mathcal{L}$ a set of expressions in propositional logic in minimal form. We call L_1 *logically distinguishable* over truth from L_2 if $\mathcal{V}(L_1 \wedge L_2) = T$ ♠

Using the above definitions for (in)distinguishability let's examine the expression $\langle X|[\Leftrightarrow]|Y\rangle$. There are two equivalent expressions, in DNF: $(XY) \vee (\neg X \neg Y)$ and in CNF: $(\neg X \vee Y) \wedge (X \vee \neg Y)$. Starting with the former which is equivalent to $\langle X|[\Uparrow]|Y\rangle + \langle X|[\Downarrow]|Y\rangle$ let $\mathcal{V}((XY) \vee (\neg X \neg Y)) = T$. This means that either XY or $\neg X \neg Y$ (or both) expressions must have a true valuation. This means that if truth is to be measured: $\mathcal{V}(X \wedge Y) \vee \mathcal{V}(\neg X \wedge \neg Y)$ which means that while X and Y are distinguishable as are $\neg X$ and $\neg Y$, we don't have that their products XY and $\neg X \neg Y$ are. It is interesting to note however that, for $\mathcal{V}(X \wedge Y) = T \Rightarrow \mathcal{V}(X \wedge Y) = F$. This hints at some amount of entanglement? It is interesting to note the similarity between $|X\rangle|Y\rangle +$

$|\neg X\rangle|\neg Y\rangle$ and $|1\rangle|1\rangle + |0\rangle|0\rangle$ where $|1\rangle$ and $|0\rangle$ are the standard basis.

In the second case, that in which $\langle X|[\Leftrightarrow]|Y\rangle = \langle X|[\Rightarrow]|Y\rangle + \langle X|[\Leftarrow]|Y\rangle$ we let $\mathcal{V}((\neg X \vee Y) \wedge (X \vee \neg Y)) = T$. Thus we must have $\mathcal{V}((\neg X \vee Y)) = \mathcal{V}((X \vee \neg Y)) = T$ hence we can distinguish between the two conjoined terms. However we can see that $Y \neq \neg X$ otherwise only one term would result in a true valuation. If $\mathcal{V}(Y) = T \Rightarrow \mathcal{V}(\neg X) = F \therefore ((\mathcal{V}(X) = T) \wedge (\mathcal{V}(\neg Y) = F))$ and we are left with $\mathcal{V}(X \wedge Y) = T$ likewise for $\mathcal{V}(\neg X) = T \Rightarrow \mathcal{V}(\neg Y) = T$ and $(\mathcal{V}(X) = F) \wedge (\mathcal{V}(Y) = F))$ which is equivalent to our original expression in CNF.

As we decided to use the valuation in comparison to 'true' as the basis for our logical expressions; that our valuations are expressed in relation to the truth of a statement we designate that

$$|X\rangle = \begin{bmatrix} X \cdot \mathcal{V}^T(X) \\ \neg X \cdot \mathcal{V}^T(X) \end{bmatrix} = \begin{bmatrix} X \cdot \mathcal{V}^T(X) \\ X \cdot \mathcal{V}^F(X) \end{bmatrix} = X|\mathcal{V}^T(X)\rangle \circledvee X|\mathcal{V}^F(X)\rangle$$

So if $\mathcal{V}^T(X) = 1$ that is, the valuation when $\mathcal{V}(X) = True$ gives that $\mathcal{V}^F(X) = 0$ as of course $\neg\mathcal{V}^T(X) = \neg 1 = 0 = \mathcal{V}^F(X)$. If we prepare a state to be true (or false) we multiply by the operators which correspond to collapsing the logical wavefunction to true (or false), which are the respective measurement matrices:

$$|1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, |0\rangle\langle 0| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Note that these look like the $[\wedge]$ and $[\neg\vee]$ measurement matrices! This is exactly what these operators are; however there is only one variable being operated on in each expression (X). If we prepare the logical expression, in this case the expression consisting only of X, to be in the truth state then we have:

$$[\wedge]|X\rangle = \begin{bmatrix} X \\ 0 \end{bmatrix} = X \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ or } X|1\rangle + 0|0\rangle = X|1\rangle$$

shown more explicitly:

$$[\wedge]|X\rangle = |1\rangle\langle 1|(X|1\rangle \circledvee \neg X|0\rangle) = X|1\rangle\langle 1||1\rangle \circledvee \neg X\ |1\rangle\langle 1||0\rangle = X|1\rangle$$

Where we pulled out X and $\neg X$ as they are the eigenvalues of the system.

This is similar to the case in which $|\neg X\rangle = [\neg\vee]|\neg X\rangle = \neg X\ |0\rangle$. Furthermore, can see that $|X\rangle\ \perp |\neg X\rangle$ as:

$$\langle X|\neg X\rangle = (\langle 1|X + \langle 0|\neg X)\textcircled{\wedge}(X|1\rangle + \neg X|0\rangle)$$

$$= \langle 1|[X \wedge \neg X]|1\rangle \textcircled{\vee} \langle 1|[\neg X \wedge \neg X]|1\rangle \textcircled{\vee} \langle 1|[X \wedge X]|0\rangle \textcircled{\vee} \langle 0|[\neg X \wedge X]|0\rangle$$

$$= (X \wedge \neg X)\langle 1|1\rangle \textcircled{\vee} (\neg X \wedge \neg X)\langle 0|1\rangle \textcircled{\vee} (X \wedge X)\langle 1|0\rangle \textcircled{\vee} (\neg X \wedge X)\langle 0|0\rangle$$

$$= 0 \textcircled{\wedge} 1 \textcircled{\vee} \neg X \textcircled{\wedge} 0 \textcircled{\vee} 0 \textcircled{\wedge} 1 \textcircled{\vee} X \textcircled{\wedge} 0 = 0$$

Definition *(Pure Logic State)*

A *pure logic state* is a truth table $T \in \mathbb{T}$ which has a single unique combination of valuations of logical variables logical expression whose valuation is equal to true. ♠

Let $\mathbb{L}$ be the set of logical expressions consisting of the logical variables X and Y. We examine the correspondence matrix for X. This can be created many ways, one of which is $\langle X|[[L]]|-\rangle$. Where it doesn't matter what we put in for $-$. Hence

$$|X\rangle = \langle X|[[L]]|-\rangle = \langle X|[[\wedge]]|-\rangle \vee \langle X|[[\Uparrow]]|-\rangle$$

We then have that X as it is part of a two variable system, that it is not pure! That we can have either Y or $\neg Y$ occurring simultaneously with X and have the valuation of either expression $X \vee Y$ or $X \vee \neg Y$ giving a valuation equal to true when $\mathcal{V}(X) = T$. In vector notation; $\mathcal{V}^T(\mathcal{L}) = \{\{T, T\}, \{T, F\}\}$ for $\mathcal{V}(L) = \{\mathcal{V}(X), \mathcal{V}(Y)\}$

However the expression $X \wedge \neg Y$ is pure as can be seen by the matrix:

$$\langle X|[\Uparrow]|Y\rangle = \langle X| \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} |Y\rangle$$

This matrix and hence the logical expression $X \wedge \neg Y$ can not be further decomposed.

Now let $\mathbb{L}$, the set of logical expressions, consist of a single logical variable X. Hence the only logical expression $\mathcal{L}$ (in minimal form) which can be created is that of $\mathcal{L} = X$. We then have that $\mathcal{V}^T(\mathcal{L}) = \{T\}$ for the ordering $\{\mathcal{V}(X)\}$ thus the logical expression X is pure when considered in a system by itself!

This gives some very interesting insight to the consideration of logical variables (or logical expressions) in light of other variables (or logical expressions). We have that a logical expression takes on context and is distinguishable only in contrast to other logical expressions! It is a worthwhile investigation in to the collapse of the wavefunction and the possibility of interference between the superpositions of different variables; that due to the measurement in respect to other variables we have a collapse of the wavefunction to states that we would not get if measured in a restricted domain, in short 'the superposition of two different variables interfere with eachother'.

Letting X be a literal, we define:

$$|X\rangle = \begin{bmatrix} X \\ \neg X \end{bmatrix}, |\neg X\rangle = \begin{bmatrix} \neg X \\ X \end{bmatrix}$$

where,

$$|X\rangle = X|1\rangle \circledvee \neg X|0\rangle$$

Things change a bit however when we go from a single logical literal to that of a logical expression. Let $X, Y \in \mathbb{L}$ and let $\mathcal{L} = X\phi Y$ where ϕ is any logical operator. We define:

$$|\mathcal{L}\rangle = \begin{bmatrix} X\phi Y \\ \neg X\phi\neg Y \end{bmatrix}, |\neg\mathcal{L}\rangle = \begin{bmatrix} \neg X\phi\neg Y \\ X\phi Y \end{bmatrix}$$

where,

$$|\mathcal{L}\rangle = X\phi Y|1\rangle \circledvee \neg X\phi\neg Y|0\rangle = \begin{bmatrix} X\phi Y \\ \neg X\phi\neg Y \end{bmatrix}$$

Looking at the projection of a Quantum-logical expression on to itself:

$$|X\phi Y\rangle\langle X\phi Y|\ |X\phi Y\rangle = \begin{bmatrix} X\phi Y & 0 \\ 0 & \neg X\phi\neg Y \end{bmatrix} \begin{bmatrix} X\phi Y \\ \neg X\phi\neg Y \end{bmatrix} = |X\phi Y\rangle$$

Now looking at a couple examples in which two logical operators ϕ and θ are different

Example 1.9 *The projection of $|X \vee Y\rangle$ on to $|X \wedge Y\rangle\langle X \wedge Y|$. We have that as:*

$$|X \wedge Y\rangle\langle X \wedge Y| = \begin{bmatrix} X \wedge Y & 0 \\ 0 & \neg X \wedge \neg Y \end{bmatrix}$$

that

$$\begin{bmatrix} X \wedge Y & 0 \\ 0 & \neg X \wedge \neg Y \end{bmatrix} \begin{bmatrix} X \vee Y \\ \neg X \vee \neg Y \end{bmatrix} = |X \wedge Y\rangle$$

so

$$|X \wedge Y\rangle\langle X \wedge Y|\ |X \vee Y\rangle\ = |X \wedge Y\rangle$$

Which is what we would expect as we can see from the matrices themselves that $[\vee] = [\Updownarrow] + [\wedge]$ *and so we would expect the restriction of* $X \vee Y$ *to* $X \wedge Y$ *to be* $X \wedge Y$ *as* $(X \vee Y) \wedge (X \wedge Y) = X \wedge Y$*).* ⧫

For an example that has one of the logical operators being asymmetric:

Example 1.10 *The projection of* $|X \Uparrow Y\rangle$ *projected on to* $|X \wedge Y\rangle\langle X \wedge Y|$*. Before we do the computation though, note that the the direct AND of* $[\wedge]$ *and* $[\Uparrow]$ *is 0 i.e.* $[\wedge] \circledwedge [\Uparrow] = [0]$*.*

$$\begin{bmatrix} X \wedge Y & 0 \\ 0 & \neg X \wedge \neg Y \end{bmatrix} \begin{bmatrix} X \wedge \neg Y \\ \neg X \wedge Y \end{bmatrix} = |0\rangle$$

Thus we have that:

$$|X \wedge Y\rangle\langle X \wedge Y|\ |X \Uparrow Y\rangle\ = |0\rangle$$ ⧫

Example 1.11 *The projection of* $|X\rangle\circledvee\ |Y\rangle$ *on to* $|X\rangle\langle X|$ *where* L *is the 'left operator' as seen in the figure 2 for the Correspondence Matrices.*

$$|X\rangle\langle X|\ |X\rangle\circledvee\ |Y\rangle\ = |X\rangle\langle X|\ |X\rangle\circledvee\ |X\rangle\langle X||Y\rangle\ = \begin{bmatrix} X & 0 \\ 0 & \neg X \end{bmatrix} \begin{bmatrix} X \vee Y \\ \neg X \vee \neg Y \end{bmatrix} = |X\rangle$$

So we have 'picked' out a particular element of the quantum-logical OR of elements. ⧫

We will later use the superposition of logical expressions (by $\circledvee$ instead of $+$) and show how they connect with faces of logical complexes.

The following is useful for looking at logical projections.

If $\langle X|Y\rangle = X \Leftrightarrow Y$ then $\mathcal{V}(X) = \mathcal{V}(Y)$, it is not necessarily that $X = Y$ as if it were then $X \Rightarrow X$ $X \vee \neg X = 1$. When $\mathcal{L}_1 = \mathcal{L}_2$ we are constraining both logical expressions to eachother. Let $\mathcal{L}_1, \mathcal{L}_2 \in \mathbb{L}$. We then have that

$\langle \mathcal{L}_1 | \mathcal{L}_2 \rangle = T$ if $\mathcal{V}(\mathcal{L}_1) = 0 = \mathcal{V}(\mathcal{L}_2)$ or $\mathcal{V}(\mathcal{L}_1) = 1 = \mathcal{V}(\mathcal{L}_2)$.

The logical vector corresponding to $X \wedge Z = [0,0,0,0,0,1,0,1]$ and that of $\neg X \wedge \neg Y = [1,1,0,0,0,0,0,0]$ and multiplication and addition between vectors is component wise.

$$\langle X \vee Y | X \wedge Z \rangle = [1,1,0,0,0,1,0,1] = X \wedge Z \vee \neg X \wedge \neg Y$$

Consider the state vector $|X \vee Y\rangle$, when projected on to itself we find: $\langle X \vee Y | X \vee Y \rangle = [0,0,1,1,1,1,1,1] + [1,1,0,0,0,0,0,0] = True$

We can make an operator by taking the outer product:

$$|X \vee Y\rangle\langle X \vee Y| = \begin{bmatrix} X \vee Y & 0 \\ 0 & \neg(X \vee Y) \end{bmatrix}$$

X	$\neg X$	Y	$\neg Y$	$X \wedge Y$	$\neg X \vee \neg Y$
0	1	0	1	0	1
0	1	1	0	0	1
1	0	0	1	0	1
1	0	1	0	1	0

Thus, for all valuations of X and Y we have that $|X \wedge Y\rangle = |X \wedge Y\rangle \veebar |\neg X \vee \neg Y\rangle$ where

$$|X \wedge Y\rangle \perp |\neg X \vee \neg Y\rangle$$

2 The Mind

Freud: How are we to arrive at a knowledge of the unconscious? It is of course only as something conscious that we know it, after it has undergone transformation or translation into something conscious [Freud 1915 P. 166]

Motivation: As of yet there is no known mechanism for 'creating' consciousness is or how it is derived from the unconscious , or even how the unconsciousness functions and where the 'boundaries' between the conscious and unconscious abide.

Here we examine more mathematically ideas of Matte-Blanco concerning the conscious and unconscious, but attempt to put these ideas on a deeper

mathematical foundation; that is, by mathematical definitions for asymmetry and symmetry. Even more interestingly, how consciousness might arise in nature.

We start by developing the notion of the 'Mind'. We then move to developing notions of a structure that we will 'concepts' as they will abide in both the 'conscious and 'unconscious'. We then go on to develop a model that is analogous to Matte-Blanco's 'Unfolding Function' by creating a 'distinguishing operator' which distinguishes asymmetry from symmetry in that it breaks apart symmetric logical expressions so one can investigate the inner workings of particular logical expressions. We then give models for the unconscious and the conscious.

If the mind can not take in the whole of what it intends to 'process' at once, but only a part is measured at a time, then there is a localization. We could possibly view the unconscious as a symmetric wave and if we have two concepts X and Y and hence $X \wedge Y$, a higher order concept. Note that this relation is the part of $X \Leftrightarrow Y$ in which both concepts are 'true' or that they both exist. We however have that X may not be distinguishable from Y. To distinguish we would make a 'measurement' in which we would collapse the higher order concept $X \wedge Y$ to that of $X \Updownarrow Y$ and distinguishing X from Y.

"(in a human or other conscious being) the element, part, substance, or process that reasons, thinks, feels, wills, perceives, judges, etc." and in Psychology: "the totality of conscious and unconscious mental processes and activities."

"judges" for example are indicative of relations. There is inherently some amounts of distinguishing between that which is judged. In the definition relating to Psychology, if the "Mind" consists of both the conscious and unconscious, then the Mind consists, at least in part, of something which has the ability to distinguish. We will say then that as the conscious has the ability to distinguish and the mind consists of the conscious, that the mind has the ability to distinguish.

Definition *(Mind)*

The *mind* consists of the Unconscious as well as the Conscious. ♠

Definition *(Mental Space)*

A *mental space* is the representation of the Mind via a set of points endowed with a topology. ♠

2.1 Concepts

Motivation: As we try to 'topologize' the mind, we look for objects of the mind in which we can represent information. Furthermore we are interested in how this 'information' is related.

In this section we proceed to define 'concepts' as well as define the 'category of concepts'. We will also investigate a hierarchical way in which to look at concepts. While we modify this idea a little for our purposes, it is an idea due to Khrennkov.

For our purposes in this thesis we let the points of the Mental Space in which we work to be 'Concepts'. We attempt to build the notion of a concept.

Definition *(Localization)*

Let S be the universe consisting of a set of points. We call *localization* of S a set of points $R \subseteq S$. ♠

"The system Ucs. treats an individual thing (person, object, concept) as if it were a member or element of a set or class which contains other members; it treats this class as a subclass of a more general class, and this more general class as the subclass or subset of a still more general class, and so on."

and as an example he states "Classes are distinguishable from one another" and that "the class of fathers are not the same as the class of mothers". Page 104: Classes are differentiated means that there are asymmetric relations that differentiate them. A class if and where the principle of symmetry can be considered as a 'portion of symmetrical being' surrounded by a layer of skin of asymmetry. This symmetry encased in asymmetry relates to a class in our unconscious.

We take these 'classes' to be 'concepts'.

Definition *(Concept)* A concept is a connected simplicial complex in which each 0-simplex (point) is connected to every other point by at least a single 1-simplex (line), but does not have more than a single line connecting any two points. Each point is an element in the category of concepts **Con**.

We view a concept as the connection between other concepts and we relate the notion of a concept to that of a simplicial complex. As complete simplicial complexes are Hausdorff (as well as the non-complete or just simplicial complexes) we have that concepts are Hausdorff as well. This can be seen as each concept has a neighborhood which is distinguishable (therefore separated) from other neighborhoods. This is true as all of the singleton concepts (simplicial complex with a single point) are both open and closed and thus are neighborhoods of themselves and therefore disjoint from other singleton concepts.

Definition *(Category of Concepts)*

We will denote **Con** as the category of concepts such that for $C_1, C_2 \in$ **Con** where C_1 and C_2 are m and n order concepts respectively $m \leq n$ and: $Hom_{\mathbf{Con}}(C_1, C_2) = \begin{cases} \{pt\} & C_1 \subset C_2 \\ \emptyset & \text{otherwise} \end{cases}$ ♠

As other categories, the *Category of Concepts* which we denote by **Con** has both objects and morphisms. We let the objects of **Con** be concepts and moprhisms be inclusions. We start by investigating the relationship between the category **Con** and **Top**. We know that **Top** is a concrete category and thus all objects are sets with additional structure; the additional structure in the case of **Top** are topologies.

We let F be a forgetful functor F : **Top** $\rightarrow$ **Set** in which assigns to each topological space the underlying set and to each continous map the underlying function.

Definition *(Related)*

Two concepts are said to be *related* if they have a point in common. ♠

If points can be distinguished, then for concepts C_1, C_2, and C_3 we have an n-cycle $C_1 \longleftrightarrow C_2 \longleftrightarrow C_3 \longleftrightarrow C_1$ as each concept is connected with each other concept. But if we relax this condition that points in common give an indiscrete set in which points are indistinguishable, then we could check for breaking of cycles which would give us asymmetries. This leads to the possibility that cycles (0-boundaries) on the simplices indicate symmetries and non-zero boundaries represent asymmetries.

Concepts are the realization of thoughts (hence they are brought in to consciousness). As Matte-Blanco mentions, the mind has to take in information piece by piece, hence in an asymmetric fashion. The result could be likened to the analogy such of that as Freud's "tip of the Iceberg" in which part of the information lies in the subconscious, which is the ice below the water, and the tip of the iceberg lies consciousness. It is important to note that in this analogy the iceberg is connected and hence while only some information is apparent in the conscious, there is a direct connection to that information lying in the unconscious. In other words all information in the unconscious does not have an equal chance of being accessed in the consciousness as analysis happens. In this scenario some information has no chance of being accessed, namely that information which is not connected in the simplex.

We will employ a similar idea to that of Khrennikov. In his paper he describes categories of order k. It is useful for our purposes in our development of a "concept". While Khrennikov, in his paper, describes the following method for his "Model of Cognitive Psychology", his use of the categories which he describes are different from our use of them. Besides the difference in the way which we are using his mental points (as points creating a hierarchy of concepts vs. ideas) another difference is that the order in which the mental points are represented is important. For the sake of simplicity we ignore the ordering. This can be seen as the forgetful functor which maps what we will define to be the concrete category of concepts to that of unordered sets.

Definition *(Concepts of Order k)*

Let $s_0, ..., s_{k-1} \in \{0, 1, ..., m-1\}$ and let the set

$$A_{s_0 s_1 \cdots s_{k-1}} = \{x = (\alpha_0, \cdots, \alpha_k, \cdots)\} \in \mathbb{Z}_m : \alpha_0 = s_0, \alpha_1 = s_1, \cdots, \alpha_{k-1} = s_{k-1}$$

We call a *concept of order k* is determined by fixing k properties". ♠

The values in this set are metric balls of radius $\frac{1}{m^k}$

Example 2.1 *First order concept: So we could, for example, take the ball $U_{\frac{1}{2^1}}$ in which we have binary choices for each order of the ball/concept. We can take a representation of this definition of a first order idea as follows: $U_{\frac{1}{2}} = x = \alpha_0\alpha_1...\alpha_N... : \alpha_0 = s_0$ Where $s_0 \in \{0, 1\}$ as it is a binary choice.* ♦

Example 2.2 *Second order concept: If we had the concept order 2 i.e. $U_{\frac{1}{2^2}}$ as opposed to $k = 1$ for the first order example. The representation of a 2nd order 2-adic (binary) concept would be*

$$U_{\frac{1}{2^2}} = x = \alpha_0\alpha_1...\alpha_N... : \alpha_0 = s_0, \alpha_1 = s_1; s_0, s_1 \in \{0, 1\}$$

♦

Example 2.3 *If we had two different concepts (Khrennikov calls them "ideas"):*

$$A_{\frac{1}{4}} = x = \alpha_0\alpha_1...\alpha_N... : \alpha_0 = 0, \alpha_1 = 1$$
$$B_{\frac{1}{4}} = x = \alpha_0\alpha_1...\alpha_N... : \alpha_0 = 1, \alpha_1 = 1$$

then the concept formed from the union of the two ball categories: $C = A_{\frac{1}{4}} \cup B_{\frac{1}{4}}$ would give a generalization of the most general order $\alpha_0 = \emptyset$ as in $C, \alpha_0 = 0$ and $\alpha_0 = 1$ therefore as $\{\alpha_0 = 0\} \cap \{\alpha_0 = 1\} = \emptyset$ we have:

$$C_{\frac{1}{2}} = \{\alpha_0 = 1\}$$

where we purposely rename $\alpha_1 \in A$ and B to $\alpha_0 \in C$. We remember that α_n is just a label. Furthermore, as there is only one attribute, C is then represented by the binary p-adic ball $C_{\frac{1}{2^1}}$.

Khrennikov however seems to get rid of α_0 in the case that it is equal to both values in this binary case. ♦

We would like to look at the categories of order k and the relation to the concept of posets. - similar to S^k? So a k-dimensional sphere which is a k-dimensional simplex. Khrennikov's first order category: Let $s \in \{0, 1, ...m-1\}$. A set $A_s = \{x = (\alpha_0, ...\alpha_k, ...) \in \mathbb{Z}_m : \alpha_0 = s\}$ which can be represented as the ball of radius $r = 1/m^1$ and any point a haveing a_0 as the first digit can be chosen as a center of this ball: $A_s = U_{1/m}(a)$, where $a = (a_0, a_1, ...), a_0 = s$ Khrennikov categories of higher orders are similar and letting $s_0, ..., s_{k-1} \in \{0, 1, ...m-1\}$. The set $A_{s_0,...,s_{k-1}} = \{x = (\alpha_0, ...\alpha_k, ...) \in \mathbb{Z}_m : \alpha_0 = s_0, ..., \alpha - 1 = s_{k-1}\}$ is a category of order k which are the balls of radius $r = 1/m_k$ Similar to Khrennikov's construction of classes The following classification example, similar to that found in Khrennikov demonstrates a classification hierarchy.

Example 2.4 *Following his example concerning the representation of a person object:* $\alpha_0 = 1$ *represents the property of being female and* α_0 *represents being male,* $\alpha_1 = 1$ *the property of being young and* $\alpha_1 = 0$ *that of being old (if there is no middle ground), and* $\alpha_2 = 1$ *for dark hair and* $\alpha_2 = 0$ *for light hair.*

The ball $U_{1/2^1} = \{x = \alpha_0\alpha_1...\alpha_N... : \alpha_0 = 1\}$

The ball $U_{1/2^2} = \{x = \alpha_0\alpha_1...\alpha_N... : \alpha_0 = 1, \alpha_1 = 1\}$

The ball $U_{1/2^3} = \{x = \alpha_0\alpha_1...\alpha_N... : \alpha_0 = 1, \alpha_1 = 1, \alpha_2 = 1\}$

The ball $V_{1/2^2} = \{x = \alpha_0\alpha_1...\alpha_N... : \alpha_0 = 0, \alpha_1 = 1\}$

$\therefore U_{1/2^3} \cap V_{1/2^2} = \{\alpha_0 = 1, \alpha_1 = 1, \alpha_2 = 1\} \cap \{\alpha_0 = 0, \alpha_1 = 1\} = \{\alpha_1 = 1\}$

Which is the the category of being old. ♦

2.2 Symmetry to Asymmetry: Blancos Unfolding Function

Motivation: There are several things that Matte-Blanco mentions in his book concerning asymmetries, one of which is the 'unfolding function'. He talks this 'unfolding' and explains how it transitions between the unconscious and conscious.

In this section we discuss the attributes of this unfolding function and expound on its meaning.

In his book Matte-Blanco states: "The most that human consciousness can achieve is to let the infinite dimensions of symmetrical being enter consciousness successively". We might imagine a door from the conscious to the unconscious where concepts are lined up in the unconscious just waiting to be pulled in to consciousness. The door, having a finite area can only allow in so many concepts simultaneously or a concept which can fit through the door. This is similar to the picture of a data bus connecting the CPU of a computer to the RAM, if there is a single line data bus which can handle a single bit, then to send through a byte means that we would have to send some bits before others. Like the CPU the conscious would have to prioritize in which order to retrieve the information from the unconscious.

Thus we look for a way for the conscious to unfold the information retrieved from the unconscious, but furthermore an ordering in which to bring in the data. This latter part could possibly be done by a density matrix operator, such as that used in quantum mechanics in which probabilities of collapsing to particular states are embedded in to this operator. We can see from our own everyday experiences that we converge to solutions i.e. pouring water in to a cup to drink works better then drinking from a flat plate hence we, by learning, converge to a solution concerning the algorithm to efficiently drink water. This is modeled by neural networks in that given training, at various epochs the network will "learn" how to do one particular task. It is possible that this density operator is derived from the minds convergence on a solution. Then when applied to a superposition of concepts, some concepts (or sub-concepts) are chosen to go through the door to consciousness before others.

We however focus on a possible method for unfolding based on boundary operations on a logical simplicial complex. An interesting area for future study would be how concepts (simplicial complexes) are chosen, but we are currently concerned in how the conscious might be able to distinguish concepts one from another and not the probability that one concept could be chosen over another or prioritized in retrieval by the conscious. For purposes of this paper we will say that consciousness is conscious of a concept that has entered it. Furthermore that there is awareness if and only if there is consciousness. It is interesting to note that self-awareness is the concept of ones-self which enters the conscious.

As Matte-Blanco states: "Words, abstract things, fulfill the function of differentiating between concepts and other things - they then are highly asymmetrical in structure." Further that "symmetrical being is translated into asymmetrical terms by the means of words (their meaning)." and that "words meaning are asymmetrical tools of the translating - unfolding function." The meaning of words causes concepts to be distinguished.

The state of being unconscious a consequence of the nature of consciousness which can not contain within itself the symmetrical being.

Page 113. What a normal man does amounts to a translating or unfolding function, consisting of the introduction of further asymmetrical relations in the symmetrical-asymmetrical 'compound'.

Consciousness can only contain highly 'asymmetrised' symmetrical pieces. When symmetry becomes greater, it becomes, for consciousness, the mute realm of the unknown.

According to Matte-Blanco, if we were to "bring in to consciousness all that there is in the 'bag' of an unconscious symmetrical being, we would have to bring in to consciousness an infinite number of things", that if "symmetrical thinking 'reads within' symmetrical being, it finds an infinite number of discrete potentialities in each class.". That the class would "unfold before the analysis of asymmetrical thinking". Hence he gives the name 'unfolding function' which we will come to discuss using the decomposition of simplices.

Between the symmetrical and asymmetrical being there is an insurmountable barrier. Page 151: Symmetry works in an "all or nothing manner", that is in jumps! Note the relationship between a continuous field and the collapse to a point.

-Symmetrical being is the normal state of man - it is the collosal base from which consciousness or asymmetrical being emerges.

- In symmetrical modes there can not be consciousness in Humans.

-Freud: Logical rules carry no weight in the unconscious.

:Concerning the negation of points - that they do not exist, Matte-Blanco states: - p is filled with p and is inconceivable without p

Words unveil meaning or 'unfolding' (distinguishing) for example, 'kitten' from 'cat' as kitten infers that it is a young cat. This is an example of how words give asymmetry as kitten $\neq$ cat, but instead kitten $\subseteq$ cat. We have that kitten $\Rightarrow$ cat, but cat $\not\Rightarrow$ kitten, in symbols:

$$U_{\frac{1}{2^1}} \Rightarrow U_{\frac{1}{2^2}} \wedge \neg(U_{\frac{1}{2^2}} \Rightarrow U_{\frac{1}{2^1}}) = \neg U_{\frac{1}{2^1}} \wedge U_{\frac{1}{2^2}}$$

thus we can have a cat but not a kitten. So it is by words that we go from $U_{\frac{1}{2^1}} = \{\alpha_0 = 1\}$ the first order category of a cat, to that of $U_{\frac{1}{2^2}}$ of being a young cat (the order of labeling the attributes is arbitrary).

2.3 The Distinguishing Operator ∂

Motivation: The 'unfolding function' which Matte-Blanco describes gives an interesting transition from the unconscious to the conscious. We attempt to model this unfolding between particular logical expressions in the unconscious as to obtain an in depth look at those in the conscious.

In this section we develop a model for the unfolding operator, namely the 'distinguishing operator' ∂. The ∂ operator is meant to both be an operator which works both as a "measurement" which breaks symmetries such as that in a quantum measurement as well as that used in the decomposition of a simplicial complex. In the quantum world measuring a particular face of a complex would represents a single possibility. The ∂ operator shows the "symmetry breaking" behavior in going from the symmetric implication $\Leftrightarrow$ to that of the breaking of this symmetry down to $\Uparrow$ or $\Downarrow$. But even deeper we look at the possibility for the minds ability to distinguish between different elements or concepts.

Definition *(Unfolding Operator of a Logical Expression)*

Let $\delta = [v_0, v_1, v_2, ..., v_n]$ be an n-simplex with vertices $v_0, \cdots v_n$, then we define the *unfolding operator of a logical expression* written ∂ to be that which acts on a logical simplex (see section Δ and gives

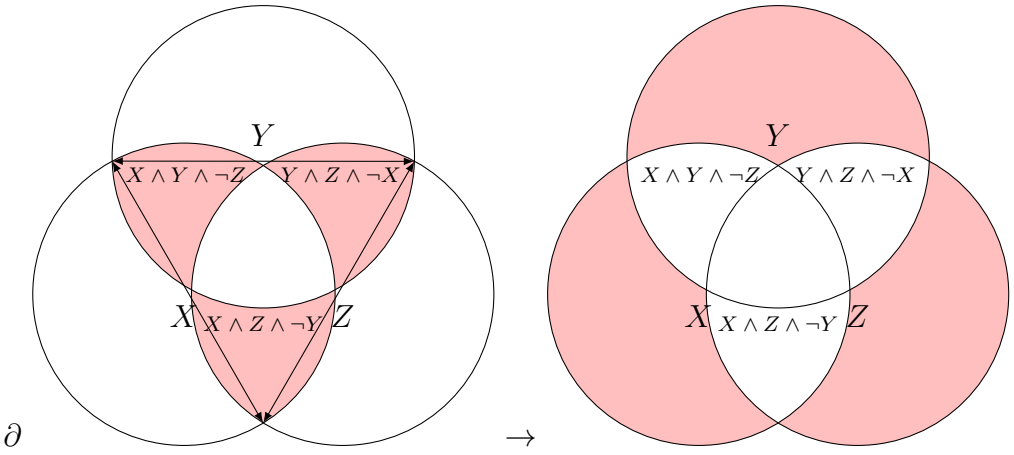
$$\partial\Delta = \bigvee_{j=0}^{n} \neg v_j \wedge [v_0, ..., v_{j-1}, v_{j+1}, ..., v_n]$$

That is, the disjunction of faces of Δ in which, for the logical expression $[v_0, ..., v_{j-1}, v_{j+1}, ..., v_n]$ comprising each face, a unique point v_j is negated. We take $\partial_v(\neg v) = 0$. We will denote the application of *partial* n-times by $\partial^n\spadesuit$

In other words, to have a logical simplicial complex, we have a particular connection between logical variables (points). When we decompose this simplex to its constituent faces we negate the truth value of one particular point, this point is different for each face of the simplicial complex. By doing this, as we shall see, end up with a set of logical expressions which when glued together create the higher dimensional simplex.

A disjunction of 3 elements gives a union of a three 0-simplices X, Y, and Z, three 1-simplices $X \wedge Y, X \wedge Z$ and $Y \wedge Z$, and a single 2-simplex $X \wedge Y \wedge Z$.

Example 2.5 *Let $X \wedge Y \wedge Z = C$ then $\partial(C) = \partial_X(C) \vee \partial_Y(C) \vee \partial_Z(C)$ which equals $(\neg X \wedge Y \wedge Z) \vee (\neg Y \wedge X \wedge Z) \vee (\neg Z \wedge X \wedge Y)$. This is precisely the three 1-simplices $X \wedge Y, Y \wedge Z$ and $X \wedge Z$ where these simplices are analogous to the lines between two distinct logical variables.*



In the above example conjunctions are connections such that, for example $X \wedge Y$ has a connection between X and Y. This conjunction is signified by the shaded regions as well as points where the tips of the arrows meet. The lines connecting these points are the disjunctions between the points in which they connect. For example, the line connecting the intersection

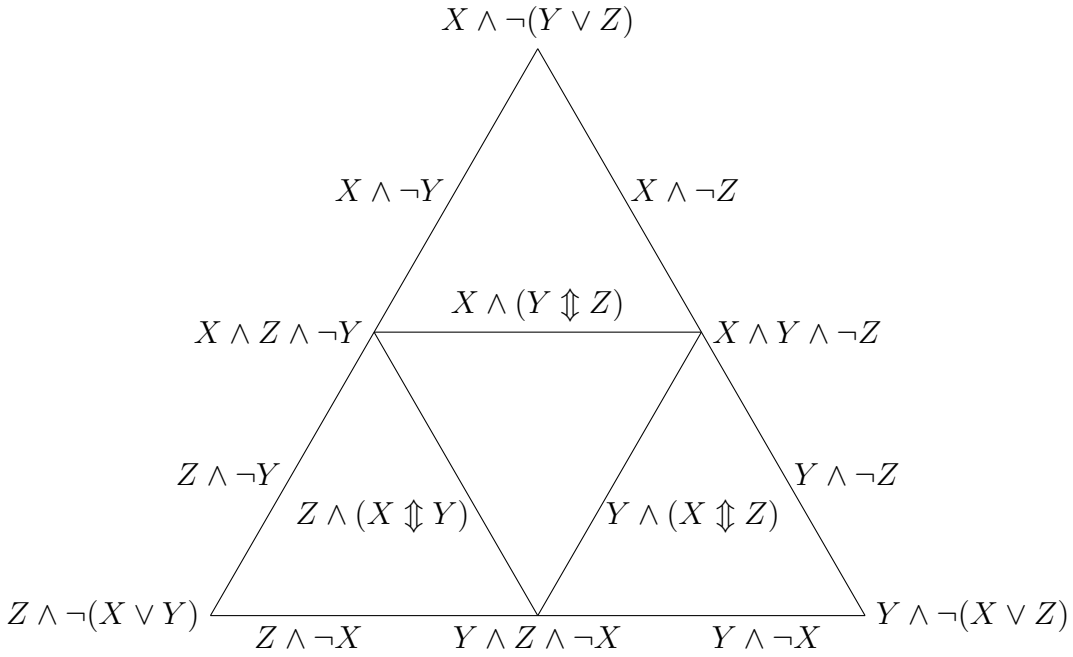


Figure 3: The unfolding of a logical triangle to constituent logical triangles.

of arrows signifying $X \wedge Y$ and $Y \wedge Z$ is the side of the with the equation $X \wedge Y \vee Y \wedge Z$.

It is both interesting and useful to note that for the 3-simplex $\Delta = v_1v_2v_3v_4$ that $\partial^3(v_1v_2v_3v_4)$ is equal to all of the single points: $\bigvee v_i \wedge \neg v_j \wedge \neg v_k \wedge \neg v_l \, \delta_{ijkl}$ where $i, j, k, l \in \{1, 2, 3, 4\}$.

Example 2.6 *The unfolding function:* $\partial^2(XYZ) = X \wedge \neg(Y \wedge Z) \vee Y \wedge \neg(X \wedge Z) \vee Z \wedge \neg(X \wedge Y)$

The reason that we go from representing a logical expression in a basis of two variables to that of 3 is the fact that, for example $X \wedge Y = X \wedge Y \wedge Z \vee X \wedge Y \wedge \neg Z$ Here we have a contraction of 3 variables to that of two (e.g. X and Y). Consider the concept of a 'green box'. Thus we have the concept of the color green, and that of a box, hence green $\wedge$ box which in itself is a concept and indistinguishable in the unconscious. However as we unfold green box via consciousness we get $\partial(\text{green} \wedge \text{box}) = \text{green} \Updownarrow \text{box}$ which allows for the possibility the concept of being green being independent of that of the concept of a box. To allow this independence it is assumed that

the mind makes allowment for another dimension, one in which the concept of green can be kept independent of that of the concept of a box.

Looking at our example in which we had the possibility:

$$\partial_X(\langle X|[\Diamond]|Y\rangle \wedge \langle Y|[\Diamond]|X\rangle) + \partial_Y(\langle X|[\Diamond]|Y\rangle \wedge \langle Y|[\Diamond]|X\rangle)$$

which is really just $\partial_X(True) \vee \partial_Y(True)$. We however can write this in a form which will shed some light:

$$\partial((X \Rightarrow Y) \vee (Y \Rightarrow X))$$

$$= \partial_X((X \Rightarrow Y) \vee (Y \Rightarrow X)) \vee \partial_Y((X \Rightarrow Y) \vee (Y \Rightarrow X))$$

In which

$$\neg X \wedge ((X \Rightarrow Y) \vee (Y \Rightarrow X)) \vee \neg Y \wedge ((X \Rightarrow Y) \vee (Y \Rightarrow X))$$

We are left with:

$$\neg X \vee \neg Y$$

This of course is $\neg(X \wedge Y)$ which means that we have unfolded from a Tautology to that of either $\neg X$ being true or $\neg Y$ being true, but not both, (or neither), but it false that both X and Y are true which leaves that if X or Y exist (valuation of true) then they have been distinguished from one another! Hence:

$$X \Updownarrow Y \vee \neg(X \wedge Y)$$

Note the similarities with the quantum like collapse of the logical expressions $(X \Rightarrow Y) \vee (Y \Rightarrow X)$

2.3.1 Gluing

Motivation: As we 'unfolded' logical expressions from a higher dimensional space to the disjunction of expressions in a lower dimensional space, we would like to build these expressions back up.

In this section we investigate the the idea of 'gluing' or 'integrating' to build up higher dimensional simplexes from lower dimensional ones as we attempt to build larger concepts from smaller ones.

Example 2.7 *Consider the unfolding of the 2-simplex (triangle) in* **??** *in to three 1-simplices (sides). We glue together the 1-simplices (sides) of the triangle by taking the conjunction of each of these sides hence:*

$$(X \wedge Y \vee Y \wedge Z) \wedge (X \wedge Z \vee Y \wedge Z) \wedge (X \wedge Y \vee Y \wedge Z)$$
$$= X \wedge Y \wedge Z$$

♦

Example 2.8 *We can build the 3-simplex (tetrahedron) by gluing together the four 2-simplex triangles which were unfolded from the original 3-simplex.*

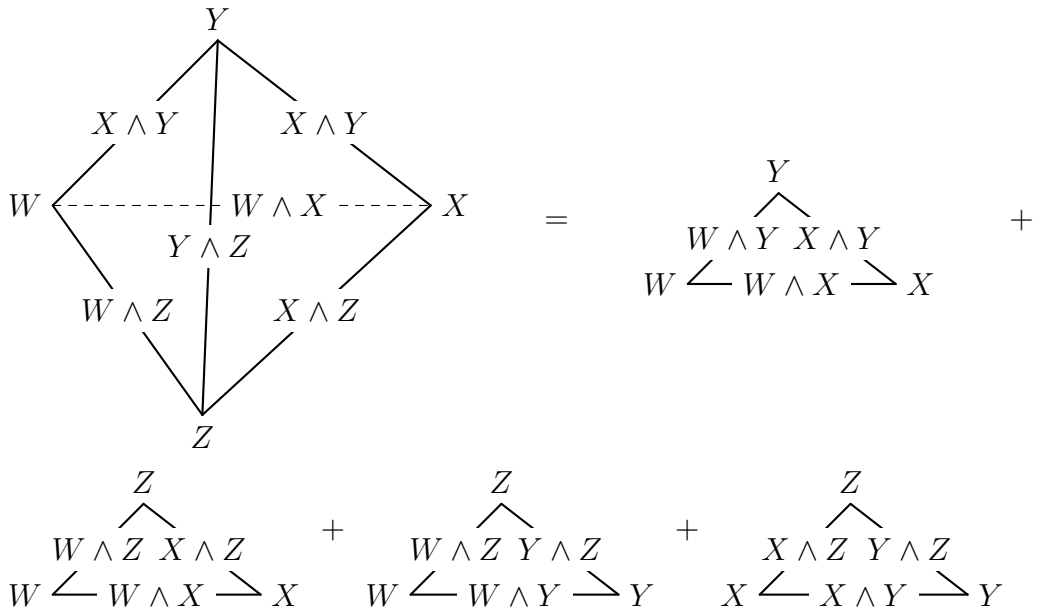


Figure 4: A decomposition of a logical 3-Simplex to the sum of logical 2-simplices

Notice the similarities between this tetrahedron and the triangle **??**. If we fixed the triangle XYZ we could possible unfold W to be the corners of the outside triangle in the above reference, that is somehow, if W could be written in terms of X,Y,and Z then:

$$\partial(W) = X \wedge \neg(Y \vee Z) \vee Y \wedge \neg(X \vee Z) \vee Z \wedge (X \vee Y)$$

That perhaps there is some kind of 'integrator' or method of gluing which builds up higher order simplices from a combination of lower order faces. A similar method is employed in CW complexes.

2.4 Unconscious

Motivation: We would like to understand how the 'unconscious' might relate to symmetry as this is a main idea of Matte-Blanco. We also would like to associate a topology to the unconscious so as to model how subsets of this topology would look and act. We might then be able to associate objects such as concepts to it.

In this section we look at various ideas in which we could create symmetry on the unconscious. We examine ideas such as that of the ultrametric principle as well as 'absence of negation' to achieve this.

Starting with a couple quotes from Matte-Blanco concerning consciousness (the corresponding pages are as superscript at the end of each quote):

"Symmetrical mode of being is all embracing and associates the individual to the class. It does not know the difference between individuals$^{[Pg.97]}$".

Furthermore he notes:

"Elements of a class are equivalent - they are however not identical." and that "The unconscious however treats them as identical", that "when we have one element before us we have the whole class$^{[Pg.139]}$"

A foundational point in Blanco's work is that he regards the Unconscious as symmetric, a main result is that symmetry causes objects to be indistinguishable from one another. This becomes important when looking at the classification of objects, if one can not distinguish between objects by their attributes, then we are able to find, for example as shown in an excerpt from Matte-Blanco:

"Let us consider the relation: y is part of x. If the converse of this relation is identical with it, that is, if the relation is symmetrical, we may say:

$$x \text{ is part of } y = y \text{ is part of } x$$

For instance, 'the arm is part of the body' is identical with 'the body is

part of the arm'. In other words, the part is identical with the whole, from which it follows logically that it is also identical with any other part. Consequently a subclass may be identical with any other subclass of the same class".

For another example, dealing with concepts, using symmetry it is possible to have the state of being alive the same as being dead. According to Matte-Blanco, the inability to distinguish between objects or concepts such as the examples given above are common to schizophrenics.

Matte-Blanco relates to the unconscious as symmetrical relationships, so for example, the unconscious would not deal with the logic of time. If it did, this would be to say that as an event X happens before event Y, symmetrically Y happens before X. Another example would be of the inability to distinguish onesself from ones parent as in a symmetric relationship to say that b is the son of a, in a symmetric relationship a would also be the son of b hence the father and son are the same.

Definition *(Unconscious)*

The *unconscious* $\mathcal{U}$ of the mental space (mind) is a collection of subsets $A, B \subseteq \mathcal{U}$, in which for $A \neq B | A \cap B = \emptyset$ ♠

Proposition 2.9 *Applying the ultrametric principle to a topological space* $(\mathcal{T}, X)$ *gives the In-discrete topology.*

Proof Let $\mathcal{T}$ be a topology on a set X and let $A \in \mathcal{T}$, $A \neq \emptyset$ and $A \neq X$. As $X \in \mathcal{T}$ by the definition of topology we have that $A \cap X \in \mathcal{T}$. But applying ultrametricity to $\mathcal{T}$ we have $A \cap X = X$ as by the ultrametric property $A \cap X \neq \emptyset \Rightarrow A \subset X$ and $X \subset A$ hence $X = A$ and we are left with the In-discrete topology. ∎

2.4.1 The Unconscious by Ultrametricity

Motivation: p-adic numbers follow the ultrametric relationship. We shall see that p-adic numbers and sets of these numbers have a relationship similar to that of unique points or ideas in categories. Furthermore these points will be developed to represent ideas or thoughts in consciousness.

2.4.1.1 Symmetry and the Unconscious

Motivation: To understand how the unconscious treats information and relationships according to Matte-Blanco we must understand symmetry.

Definition *(Symmetry)*

Let $x, y \in \mathbf{Ob}$, where **Ob** is the category of objects and xRy be a binary relationship on objects x and y. We define a relationship to be *symmetric* if $\forall x \forall y[xRy \Leftrightarrow yRx]$ ♠

Example 2.10 *Let us take the implication $xRy \Rightarrow yRx$ for any $x, y \in$ **Ob**. Applying symmetry we consider the reverse implication $yRx \Rightarrow xRy$*

$$xRy \Rightarrow yRx \wedge yRx \Rightarrow xRy$$

$$= (\neg x \vee y) \wedge (\neg y \vee x)$$

$$= (x \wedge y) \vee (\neg x \wedge \neg y)$$

$$= xRy \Leftrightarrow yRx$$

. Furthermore we have that the reflexive and transitive relationships are also satisfied. Thus we have an equivalence relationship. ♦

Lemma 2.11 *Applying symmetry, the proper subset becomes equal to the whole set.*

Proof *Let a be a proper subset of A. By symmetry $a \subset A \Rightarrow A \subset a$ Thus we have $A = a$ Hence the proper subset becomes equal to the whole set.* ∎

Example 2.12 *If we want to look at this relationship in terms of sets, if $\{5\} \subset \{5, 7\}$ then $\{5, 7\} \subset \{5\}$ hence the subset (part) is equal to the whole set.* ♦

Another interesting result, which can be seen in the alive vs. dead example is the absence of negation. This makes the law of contradiction and asymmetric logic, such as implication, invalid. This can be seen as if $p = \neg p$ where $p = (x \Rightarrow x)$:

$$(x \Rightarrow \neg x) = \neg(x \Rightarrow \neg x) = \neg x = x$$

So we can see directly that implication becomes meaningless hence if the property of being alive was associated with x and that of being dead of $\neg x$, then with absence of negation the ability to distinguish the difference between being alive and that of being dead. If each concept is represented as a p-adic ball with each of its attributes (which are concepts as well) we examine the possibility for comparison (distinguishing) of concepts if the ultrametric property is introduced.

Let U and V be the p-adic representation of concepts of order m and n respectively, that is; $U_{\frac{1}{p^m}}$ and $V_{\frac{1}{p^n}}$. Let $U \cap V = W$ where $W = W_{\frac{1}{p^k}} \neq \emptyset$. We will consider the 2-adic representation for simplicity. Let $x \in W$ then we have that $x \in U \cap V \Rightarrow x \in U$ and $x \in V$ we then have that we can relate x to a point $y \in U$ and $z \in V$. We note that although U, V and W may all be concepts of different orders, we consider the case in which attributes which are not shared between concepts have an empty intersection. This is a similar idea to a vector subspace in which the subspace can be projected on to the original vector space by appending 0's for the unshared dimensions between the subspace and the larger vector space containing it. As we are interested in the attribute which is related to both of the p-adic balls we treat the unshared attributes between the balls as 0's (or false concerning the existence of an attribute), hence they do not occur in the attribute which is shared between the balls.

We are interested in how y is related to z when considering the ultrametric.

Now employing the ultrametric principle, in particular to examine the relationship between y and z we have:

$$|y - z|_2 \leq max|x - z|_2, |x - y|_2$$

2.5 Conscious

Motivation: The motivation in this section is to show how a logical system can go from being symmetric to being asymmetric, that is, going from a state that would be akin to being in the unconscious, to a possible conscious state.

In this section we attempt to create logical asymmetries from symmetries by the collapse of a 'logic wave'. We however also look at the partial operator

∂ (the unfolding function) so as to see how it relates to a 'measurement' of a logic wave.

Turning our attention towards consciousness, we start by looking at the dictionaries definition: "Consciousness is the act or process of thinking or the act of carefully thinking about the details of something". While "carefully" depends on how much care one believes should be taken, this does imply the focusing of attention and it is in this focusing in which the details become apparent and furthermore that there is distinguishing of the details.

Matte-Blanco says:

"Consciousness can not exist without asymmetrical relations because the essence of consciousness is to distinguish and differentiate which can not be done by symmetrical relations alone.[$Pg.96$]"

As Matte-Blanco mentions, in describing words as asymmetrical and hence indicative of consciousness as to distinguish words one must have asymmetry. Consider the following statement: "I think therefore I am" Note the presumption of "I". The statement makes an presumption or assumption that "I" already exist thus distinguishing it from non-existence.

Definition *(Conscious)*

The *Conscious*$\mathcal{C}$ is a subset topology of the mental space in which subsets of $\mathcal{C}$ are separated. ♠

2.5.1 The Conscious by Wave Collapse

As long as you don't choose, everything remains possible -Mr. Nobody

Motivation: In examining the unconscious we found that by incorporating the ultrametric principle into a topology, that either the points in this topology are distinct or indistinguishable. Furthermore, that this indistinguishably of points created symmetry. As we attempt to model consciousness according to asymmetric principles we relax the principle of ultrametricity as we want to distinguish points (concepts).

Matte-Blanco states:

".. It [symmetry to asymmetry] works in an all-or-nothing manner, that is, in jumps: this would be against the hypothesis of a continuum of thought

from the asymmetrical to the symmetrical type of establishment of relations or from conscious to unconscious"[Pg.151]

We will start by defining symmetry as related to a quantum mechanical system as well since we will be dealing heavily with matrices and quantum mechanics through the rest of their paper.

Definition *(Symmetry)*

A matrix is *symmetric* if it is equal to its own transpose. More formally a matrix A is symmetric if $A = A^T$. ♠

Symmetry is an invariant over a particular transformation. In our logical expressions, if we can swap the left hand side with that of the right hand side we have symmetry. We can see that the symmetric operators then are:

$$\Updownarrow, \Leftrightarrow, \Updownarrow, \wedge, \vee, \neg\wedge, \neg\vee, \mathbf{0}$$

Definition *(Asymmetry)*

A matrix is *asymmetric* if it is not symmetric. ♠

We have the asymmetric operators as represented by matrices being:

$$[\Rightarrow], [\Leftarrow], [\Uparrow], [\Downarrow], [L], [\neg L], [R], [\neg R]$$

see figure **??**).

Proposition 2.13

The expression $\langle A|[\mathcal{G}]|B\rangle = \langle B|[\mathcal{G}]|A\rangle$ *is symmetric for* $\mathcal{G} \in \{\Leftrightarrow, \Updownarrow, \Updownarrow\}$.

Proof Since $\mathcal{G}$ is symmetric for all $\mathcal{G} \in \{\Leftrightarrow, \Updownarrow, \Updownarrow\}$ the result follows.

However a little more insightful is the following proof.

Proposition 2.14 *The* $[\Updownarrow]$ *is symmetric if both* $[\Updownarrow]$ *and* $[\Leftrightarrow]$ *are symmetric.*

Proof *It can be shown that:*

$$\langle A|[\Updownarrow]|B\rangle = \langle A|[\Leftrightarrow]|B\rangle + \langle A|[\Updownarrow]|B\rangle$$

But as

$$\langle A|[\Leftrightarrow]|B\rangle = \langle B|[\Leftrightarrow]|A\rangle \text{ and } \langle A|[\Updownarrow]|B\rangle = \langle B|[\Updownarrow]|A\rangle \text{ we have:}$$
$$\langle B|[\Updownarrow]|A\rangle + \langle B|[\Leftrightarrow]|A\rangle = \langle B|[\diamondsuit]|A\rangle$$

Now all of the matrices $\Leftrightarrow, \Updownarrow$, as well as $\diamondsuit$ are not only equal to their transpose, but they are also equal to their inverses. Note the Hermitian property! The Hermitian property is not shared the case of $[\Rightarrow], [\Leftarrow], [\Uparrow]$, or $[\Downarrow]$ and that these latter operators are asymmetric. This will be extremely important when investigating a model for symmetries and asymmetries in the unconscious and conscious respectively.

Note Note that in the case of $X \Leftrightarrow Y = (\neg X \vee\ Y) \wedge (\neg Y \vee X)$ we can just flip the AND's to OR's and OR's to AND's and get the same thing. We can not do this in the case of a left or right implication! $X \Rightarrow Y = \neg X \vee Y$ but flipping OR to AND gives us $\neg X \wedge Y = \neg(Y \Rightarrow X)$ Which is not the same. This may be an equivalent definition? to the symmetry of an expression.

Let us examine the $\diamondsuit$ operator a bit more in depth. Consider $\langle X|[\diamondsuit]|Y\rangle = \langle X|[\Updownarrow]|Y\rangle + \langle X|[\Leftrightarrow]|Y\rangle$ as in the above example. However now let's work out the details.

$$\langle X|[\diamondsuit]|Y\rangle = X\neg Y + \neg XY + XY + \neg X\neg Y = X(Y + \neg Y) + \neg X(Y + \neg Y) = X + \neg X = True$$

We indeed did have a superposition of $\langle X|[\Updownarrow]|Y\rangle$ and $\langle X|[\Leftrightarrow]|Y\rangle$ then we would have that the system when measured in respect to the X and $\neg X$ basis could collapse to either X or $\neg X$, that is; either X is true, or it is false. So what started off as a Tautology, when measured in respect to the X basis, we found that the system could collapse to that of either an observation of the truth or falsity of some observable! This may seem strange at first glance, but is believed that this type of system should be considered.

Having a measurement being true or false depends on something being measured and to say that a system such as a logical expression or a mass particle etc. is true or observed assume that there is something to be observed. Once this assumption is made, lack of observing it (in the case of a mass particle) or finding a contradiction in the case of a logical expression gives rise to being able to distinguish the system. That is to say, if we

$$|\text{Existence}\rangle + |\text{non-Existence}\rangle \rightarrow \text{Existence} \Updownarrow \text{non-Existence}$$

That is, upon making an observation the observer can find that something exists or does not exist, but not both (if it is indeed the case that existence and non-existence are mutually exclusive).

If it is true that a system exists then to find out if it exists a measurement must be made. This goes down the road of what is known in quantum physics as the "measurement problem" and whether or not a system exists without measurement. It may be possible however that the system exists, but an observer may not have knowledge of the system. It is similar to the famous statement made by Descartes, translated in English to, "I think, therefore I am". He however pre-supposes that he exists (by distinguishing himself and refering to himself as "I"). His self-measurement or awareness of himself (or self-awareness) led him to believe in his own existence.

Continuing down the path of observing X or not X from a Tautology on a system of two logical variables X and Y as $\langle X|[\Updownarrow]|Y\rangle = \langle Y|[\Updownarrow]|X\rangle$ we may have the possibility of observing Y or $\neg Y$ when measured in the Y basis. As $\langle X|[\Updownarrow]|Y\rangle = \langle X|[\Updownarrow]|Y\rangle \vee \langle Y|[\Updownarrow]|X\rangle$ we have that:

$$|\langle X|[\Updownarrow]|Y\rangle \vee \langle Y|[\Updownarrow]|X\rangle \rangle = (X \vee \neg X) \vee (Y \vee \neg Y)$$

Notice however that as $X \vee \neg X \vee Y \vee \neg Y = (X \vee \neg Y) \vee (Y \vee \neg X)$

$$|\langle X|[\Updownarrow]|Y\rangle \vee \langle Y|[\Updownarrow]|X\rangle \rangle = (X \Rightarrow Y) \vee (Y \Rightarrow X)$$

We have that, among the possibility for collapse of the wavefunction, we could measure outcomes which give rise to implication!

Looking at this a little differently we have:

$$|X \Updownarrow Y\rangle = |X \Leftrightarrow Y\rangle \circledcirc |X \Updownarrow Y\rangle$$

Which if there is a logic wave collapse, we may have the possibility of collapsing to $|X \Updownarrow Y\rangle$. However as nature rarely gives things in pairs which are detected simultaneously as time is involved and events happen seperately, we might measure the system in the occurance of one event X or the other Y and not both at the same time. This is again $X \Updownarrow Y$ and our projection or measurement matrix $[P]$ is then $|X \Updownarrow Y\rangle\langle X \Updownarrow Y|$ and upon measurement we get:

$$\langle X \Mapsto Y || X \Updownarrow Y \rangle \langle X \Updownarrow Y || X \Mapsto Y \rangle = X \Updownarrow Y$$

So we measure X or Y but not both. This is what we would expect as we are measuring in respect to 'truth' or, in this case, that an event is occurring.

But furthermore, that these events can be measured distinct from one another!

Q: What if we could break apart 0 such that $0 = 1 + -1$, then make a measurement which collapses it to one of the two states ex. spin up/down?

Note also that

$$\partial(X \Mapsto Y) = \partial(X \vee Y) \oslash \partial(\neg X \wedge \neg Y) = X \Updownarrow Y$$

2.5.2 ∂ Operator on Exact Sequences/Logical Expressions

The ∂ operator distinguishes the image from the kernel? The matrix in the middle could possibly serve as the "measurement operator" in that for example polarization we can measure the light causes a 'collapse' or choice of a particular state. Possibly, after we make a measurement we have a 'polarization' and if we have an exact sequence then this second matrix is perpendicular to the first (image = kernel) hence we would have a contradiction. So it is possible if the sequence is exact then there is a contradiction between subsequent mappings. The amount of contradiction (or distinguishment) can be seen from the projection operator! Note that in any set of 3 matrices they are exact only if the component wise $\wedge$ is 0 between them all. This seems like it might follow polarization in that the total polarization is negated.

So now I have to relate a measurement operator say $[\theta]$ to its projection, so $[\theta]|\phi\rangle$. Also one operators operation on another operator, say $[\theta][\phi]$ is the $\oslash$, but they constrain to cause projection and if $[\theta][\phi] = [0]$ then we possibly have a logical contradiction.

Finding the image in respect to a function (image of a function) in the correspondence matrices is like putting a constraint on the function - so, for example $[\vee] \to [\Updownarrow]$ polarizes the matrix (the logic). If we take the valuation of true and not both true and false so each image $f(\mathcal{V}_F)$'s exact if f is the XOR function?

$\partial[\wedge] = \partial[\Updownarrow] = \partial[\Uparrow] \oslash \partial[\Downarrow]$ Look at exact sequences - or how far from exactess

a sequence is and after moding out the non-exactness, if we have exactness.

If we take two logical operators $[\theta]$ and $[\phi]$ then the amount of perpendicularity is $[\theta] \oslash [\phi]$. Now we could have the case such that $[\theta] \oslash [\phi] = 0$ and $[\theta] \otimes [\phi] = [\oplus]$. So in the case of $[\wedge] \oslash [\Uparrow] = [0]$

$$|X \wedge Y\rangle = \begin{bmatrix} X \wedge Y \\ \neg(X \wedge Y) \end{bmatrix}, \langle X \wedge Y| = \begin{bmatrix} X \wedge Y & \neg(X \wedge Y) \end{bmatrix}, |X \wedge Y\rangle \langle X \wedge Y| =$$
$$\begin{bmatrix} X \wedge Y \\ \neg(X \wedge Y) \end{bmatrix} \otimes \begin{bmatrix} X \wedge Y & \neg(X \wedge Y) \end{bmatrix} = \begin{bmatrix} X \wedge Y & 0 \\ 0 & \neg(X \wedge Y) \end{bmatrix}$$

Which then looking at the projection on to $|X \wedge \neg Y\rangle$ we have:

$$|X \wedge Y\rangle \langle X \wedge Y||X \wedge \neg Y\rangle$$
$$= \begin{bmatrix} X \wedge Y & 0 \\ 0 & \neg(X \wedge Y) \end{bmatrix} \begin{bmatrix} X \wedge \neg Y \\ \neg(X \wedge \neg Y) \end{bmatrix} = \begin{bmatrix} 0 \\ X \end{bmatrix} = X|0\rangle$$

Hence we have X in the direction of falsity or $\neg X$.

Now consider logical expressions which have overlapping truth tables. Let $[\theta] = [\Rightarrow]$, $[\phi] = [\vee]$, $[\Rightarrow] \oslash [\vee] = [R]$, and $X[R]Y = Y$. As:

$$|X \Rightarrow Y\rangle \langle X \Rightarrow Y| = \begin{bmatrix} X \Rightarrow Y & 0 \\ 0 & X \Uparrow Y \end{bmatrix}$$

We have:

$$|X \Rightarrow Y\rangle \langle X \Rightarrow Y||X \vee Y\rangle = Y|1\rangle$$

Note that there is no agreement between the two logical matrices over 0 hence there is no valuation resulting in true in which the atoms in the expression are evaluated over false.

We try the general case in which we let the logical vectors be $|X[\theta]Y\rangle$ and $|X[\phi]Y\rangle$. computation shows that:

$$|X[\theta]Y\rangle \oslash |X[\phi]Y\rangle = |X[\theta \oslash \phi]Y\rangle$$

which is also the case with the transpose of each i.e.:

$$|X[\theta]^T Y\rangle \oslash |X[\phi]^T Y\rangle = |X[\theta \oslash \phi]^T Y\rangle$$

As we found in **??**: $\neg\langle X|[\varrho]|Y\rangle = \langle Y|[\neg\varrho]^T|Y\rangle$ and as:

$$\begin{aligned}\langle Y|[\neg\theta]^T|X\rangle\langle Y|[\neg\phi]^T|X\rangle &= \langle Y|[\neg\theta]^T \barwedge [\neg\phi]^T|X\rangle \\ &= \langle Y|\neg([\theta] \barvee [\phi])^T|X\rangle = \neg\langle Y|[\theta] \barvee [\phi]|X\rangle\end{aligned}$$

as $\barvee$ is commutative. We then have the general expression:

$$|X_i[\phi \barwedge \theta]^i_j Y^j\rangle = X_i[\phi \barwedge \theta]^i_j Y^j|1\rangle + X_i[\phi \barvee \theta]^i_j Y^j|0\rangle$$

in tensor notation.

3 Logical Topology and Complexes

Motivation:

3.1 Logical Topology

Motivation: Concerning consciousness Matte-Blanco goes on to say: "Consciousness in contrast when confronted by a whole class can consider it in only 2 ways - either it focuses on the limits (or defn) of the class, that is on whose precise features which characterize it and distinguish it from all other classes or it concentrates on the individuals which form the class."

This "limit" could be modeled by the Stalk (see Sheaf Theory). As one gets finer and finer and filters through various elements of a class that, while are not equivalent (thus in the conscious), it is possible that there are several Germs.

One may notice the similarities between $\vee$ and $\wedge$ and $\cup$ and $\cap$ respectively. We proceed to utilize this relationship and create a topology of logical expressions as well as structures such as sheaves and schemes in our quest to create simplicial complexes from these logical expressions. We start by relating the categories of Logical expressions **Log** to that of **Set**, where the objects of **Log** are logical expressions and the morphisms are inclusions. We then look for a functor $F : Hom_{\mathbf{Log}}(-,-) \to Hom_{\mathbf{Set}}(-,-)$ which map the inclusions in **Log** to those of inclusions in **Set**. This is the forgetful functor. We can then create presheaves from the logical expressions attached to the objects (sets) in **Set**. Furthermore we will then attempt to glue the sections of these presheaves together to create sheaves of logical expressions.

We start by taking $\mathcal{L} \in \mathbf{Log}$ in and re-writing it in DNF $\mathcal{L} = \mathcal{L}_1 \vee \mathcal{L}_2 \vee .. \vee \mathcal{L}_n$ in which we have that $\forall\, i \in \{1, 2, .., n\}, \mathcal{L}_i \in \mathbf{Log}$, each $\mathcal{L}_i$ corresponds to some $F(\mathcal{L}_i) \in \mathbf{Set}$, and also that for each $\mathcal{L}_i, \mathcal{L}_j \; \exists f \in Hom_{\mathbf{Log}}(\mathcal{L}_i, \mathcal{L}_j)$ such that there is an $F(f) \in Hom_{\mathbf{Set}}(F(\mathcal{L}_i), F(\mathcal{L}_j))$ hence $F : \mathbf{Log} \to \mathbf{Set}$ is surjective. This can be seen as for any two unique elements in **Set** one can construct two unique sentences in **Log** along with their maps.

Definition *(Category of Logical Expressions [**LogEx**])*

The *category of logical expressions (**LogEx**)* has logical expressions as objects and morphisms are finite logical transformations. ♠(possibly need these def's in respect to partitions).

Hence elements of **LogEx** are the logical expressions in $\mathbb{L}$ with finite maps which transform one truth table in to another.

Definition *(Category of Logical Sets [**LSet**]*

The *category of logical sets (**LSet**)* has as objects unordered sets consisting of logical variables and whose morphisms are (inclusions)?. ♠

If morphisms are inclusions, then we naturally have presheaves.

Example 3.1 *Let $\mathcal{L} \in$ **LogEx** Now, this depends on whether I use partitions or not.*

Definition *(Logical Set)*

A *logical set* on a logical expression $\mathcal{L} \in$ **LogEx** is a set representation of $\mathcal{L}$ via a morphism $\varsigma : \mathcal{L} \to \mathcal{S} \in$ **LSet** where **LSet** as the category of sets of logical expressions. ♠

It is important to note that the sub-expressions of literals may be composed of single logical variables. This will be the case in our logical basis transformation which results in the universal set being partitioned which in this new basis the conjunction of any two logical variable is empty.

Proof We will prove that ς is well defined on an arbitrary logical expression $\mathcal{L}$...

Definition *(Covering Set Over Truth)*

The *covering set over truth* $\mathcal{C}_{\mathcal{S}}$ of a set $\mathcal{S} = \varsigma(\mathcal{L})$, $\mathcal{L} \in$ **LogEx** is a logical set which has at least the same true valuations as $\mathcal{L}$. ♠

Example 3.2 *Let a logical expression $\mathcal{L} = \mathcal{L}_1 \vee \mathcal{L}_2$. Re-writing the expression in set notation, we have $\varsigma(\mathcal{L}) = \mathcal{S} = \{\varsigma(\mathcal{L}_1), \varsigma(\mathcal{L}_2)\}$ and as $\{\varsigma(\mathcal{L}_1)\} \cup \{\varsigma(\mathcal{L}_2)\}$ contains the set of all true valuations of $\mathcal{L}$ we have that $\varsigma(\mathcal{L})$ is covered by $\mathcal{C}_{\mathcal{S}} = \{\mathcal{L}_1, \mathcal{L}_2\}$. Since $\varsigma(\mathcal{L}) \simeq \mathcal{S}$ we have that $\mathcal{L}$ is covered by $\mathcal{L}_1 \vee \mathcal{L}_2$* ♦

This last example was a covering by construction as we took $\mathcal{L}$ to be equal to $\mathcal{L}_1 \vee \mathcal{L}_2$. This does not always have to be the case. Note for example that a covering set (over truth) for $\mathcal{S} = \mathcal{L}_1 \wedge \mathcal{L}_2$ is $\mathcal{C}_\mathcal{S} = \mathcal{L}_1 \vee \mathcal{L}_2$ as all of the true valuations for any combination of literals in $\mathcal{S}$ are true in $\mathcal{C}_\mathcal{S}$ but $\mathcal{S} \neq \mathcal{C}_\mathcal{S}$

Definition *(Open Sentence (Open Logical Expression)*

An *open sentence* or *open logical expression* is the simultaneous false valuation on each atom (non-negated logical variable) resulting in the logical expression being true. ♠

Example 3.3 *Let the logical expression $\mathcal{L} = X \Rightarrow Y$. As $\mathcal{V}_F(\mathcal{L}) = F \Rightarrow F = T$ we have that $\mathcal{L}$ is an open sentence.* ♦

Definition *(Closed Sentence (Closed Logical Expression)*

A *closed sentence* or *closed logical expression* is one in which its compliment is open. ♠

Notice that these definitions are analogous to open and closed in point set topology.

Concerning notation; we will denote the conjunction of two literals by a comma. We will denote the disjunction of two subexpressions by a comma between the brackets which enclose the sets of literals. We do this because it is more informative by looking at the expressions as sets. For an explicit example take the expression $\mathcal{L} = X \Leftrightarrow Y$. In DNF it is equivalent to the expression: $(X \wedge Y) \vee (\neg X \wedge \neg Y)$. We then turn $\mathcal{L}$ in to a logical set $\mathbf{Set}(\mathcal{L})$ by $\varsigma(\mathcal{L}) = \{\{X, Y\}, \{\neg X, \neg Y\}\}$ which would be a covering set for $\mathbf{Set}(\mathcal{L})$. It is very important to note that we could have more than one covering set, and in this example we could also have $\mathbf{Set}(\mathcal{L}) = \{\{X\}, \{Y\}, \{\neg X, \neg Y\}\}$ however this covering set also covers everything i.e. $X \vee Y \vee (\neg X \wedge \neg Y) = T$ no matter the values of X or Y! ♦

Proposition 3.4 The finite conjunction of open logical expressions is open

Proof

We can prove this by induction. Let A and B be two open logical expressions. As the false valuation for both is true we have; $\mathcal{V}_F(A) \vee \mathcal{V}_F(B) =$

$T \vee T = T$. Now let C be an open logical expression formed from the conjunction of a finite number of other open logical expressions and conjoin another open sentence D to C with the OR operation then $\mathcal{V}_F(C) \vee \mathcal{V}_F(D) = T \vee T = T$ hence proving that the finite conjunction of open logical expressions is open. ∎

Proposition 3.5 The finite disjunction of open logical expressions is open

Proof

Similar to the proof in which the finite conjunction of open expressions is open we let A and B be two open expressions, hence $\mathcal{V}_F(A) \wedge \mathcal{V}_F(B) = T$. By induction we let C be any finite disjunction of open logical expressions and conjoin another open expression D using the AND operation and we have the final expression $\mathcal{V}_F(C) \vee \mathcal{V}_F(D) = T \vee T = T$. ∎

Definition *(Propositional Logic Topological Space [ProLT])*

A *propositional logic topological space (ProLT)* denoted $\mathcal{T}$ is a topology created from the first order predicates $\neg, \wedge, \vee$ in which the $\wedge$ and $\vee$ correspond to $\bigcap$ and $\bigcup$ respectively. We will overload the notation and denote a ProLT by $\varsigma(\mathcal{L})$ the Topological space associated with the logical set $\varsigma(\mathcal{L})$ on $\mathcal{L} \in \mathbb{L}$. ♠

Example 3.6 *Consider the expression $X \Leftrightarrow Y = (X \wedge Y) \vee (\neg X \wedge \neg Y)$. This is already in CNF and we can see that there are two subexpressions $(X \wedge Y)$ and $(\neg X \wedge \neg Y)$ joined together. The notation changes to $\{\{X, Y\}, \{\neg X, \neg Y\}\}$. The logical expression in this set notation will be deemed the covering set.* ♦

Example 3.7 *Let $\varsigma(\mathcal{L}) = \{\{X\}, \{Y\}, \{\neg X\}, \{\neg Y\}\}$ 4 points which cover the whole space $\varsigma(\mathcal{L})$. Such as the discrete topology the combination of these points satisfy the criterion for a topology on the sets which we will call $\varsigma(D)$, the set of all sets of literals and expressions for each combination of literals $l_i \in \varsigma(\mathcal{L})$.*

$\varsigma(\mathcal{D}) = \{X, \{X\}, \{Y\}, \{\neg X\}, \{\neg Y\}, \{\{X\}, \{Y\}\},\ \{X, Y\}, \{\{\neg X\}, \{\neg Y\}\},$

$\{\neg X, \neg Y\}\{X, \neg Y\}, \{\{X\}, \{\neg Y\}\}, \{\neg X, Y\}, \{\{\neg X\}, \{Y\}\}, \emptyset\}$

This satisfies the criteria for a topology as both $\varsigma(\mathcal{L})$ *and* $\emptyset$ *are in* $\varsigma(\mathcal{D})$ *and for the open sets:*

$\{X, \{\neg X\}, \{\neg Y\}, \{\{\neg X\}, \{\neg Y\}\}, \{\{X\}, \{\neg Y\}\}, \{\{\neg X\}, \{Y\}\}, \{\neg X, \neg Y\}, \emptyset\}$
and both the unions and intersections of each open set is in element in this set of open sets we have that $\varsigma(\mathcal{D})$ *is indeed a topology.* ♦

Example 3.8 *We can look at the induced topology on the discrete topology* $\varsigma(\mathcal{D})$ *given by the points* $\varsigma(\mathcal{L}) = \{\{X\}, \{\neg Y\}\}$ *which gives us:*

$$\varsigma(\mathcal{D}) \mid_{\varsigma(\mathcal{L})} = \{X, \{X\}, \{\neg Y\}, \{\{X\}, \{\neg Y\}\}, \{\{X, \neg Y\}\}, \emptyset\}$$

Where $\{\neg Y\}$ *and* $\{\{X\}, \{\neg Y\}\}$ *are open. It can be checked that any* $a, b \in \varsigma(\mathcal{D}) \mid_{\varsigma(\mathcal{L})}$ *over* $\cup, \cap$ *gives back an element in* $\varsigma(\mathcal{D}) \mid_{\varsigma(\mathcal{L})}$ *hence* $\varsigma(\mathcal{D}) \mid_{\varsigma(\mathcal{L})}$ *satisfies the requirements of a topology.*

Theorem 3.9 *The open and closed sentences define a ProLT.*

Proof *Let* $\varsigma(\mathcal{L}) = L$ *denote the set obtained from the logical expression* $\mathcal{L}$ *and let* $L = L_1 \cup L_2$ *where* $L_1 = \varsigma(\mathcal{L}_1)$ *and* $L_2 = \varsigma(\mathcal{L}_2)$ *for* $\mathcal{L} = \mathcal{L}_1 \vee \mathcal{L}_2$*. Let both* L_1 *and* L_2 *be open hence they both contain FINISH!*

Before we proceed we employ the change of basis as seen in **??**. We list the transformations here:

A.	$X \wedge \neg(Y \vee Z)$
B.	$Y \wedge \neg(X \vee Z)$
C.	$Z \wedge \neg(X \vee Y)$
D.	$(X \wedge Y) \wedge \neg Z$
E.	$(X \wedge Z) \wedge \neg Y$
F.	$(Y \wedge Z) \wedge \neg X$
G.	$X \wedge Y \wedge Z$
H.	$\neg X \wedge \neg Y \wedge \neg Z$

Example 3.10 *Let us generate a topology on the logical expression* $\mathcal{L} = X \vee \neg Y$ *in the partitioned basis given above. First we note that in considering the logical expression* $\mathcal{L} = X \vee \neg Y$*, as* $\mathcal{V}^T(\mathcal{L}) = \{\{T, F\}, \{F, F\}, \{T, T\}\}$ *for the ordered pair* $\{\mathcal{V}(X), \mathcal{V}(\neg Y)\}$ *we have that a valuation can be equal to true solely on that of the valuations of either the logical points* X *or* $\neg Y$*. Thus*

we take these points as the basis for our ProLT and find that In a logical set: $\varsigma(\mathcal{L}) = \mathcal{S} = \{A, C, D, E, G, H\}$.

In the partitioned basis $X = \{A, D, E, G\}$, $\neg Y = \{A, C, E, H\}$, *and* $X \wedge \neg Y = \{A, E\}$. *We take the topology on* $\mathcal{S}$ *to be*

$$\{\mathcal{S}, \{A, D, E, G\}, \{A, E\}, \{A, C, E, H\}, \emptyset\}$$

which was derived by taking the unions and intersections of $\varsigma(X)$ *and* $\varsigma(\neg Y)$. *More explicitly we can see that*

$$\{A, D, E, G\} \cup \{A, C, E, H\} = \{A, C, D, E, G, H\} = \mathcal{S}$$

and that $X \wedge \neg Y$ *is actually derived from:*

$$\{A, D, E, G\} \cap \{A, C, E, H\} = \{A, E\}$$

Hence we have constructed a ProLT which relates the vectors $X \vee \neg Y$*; rewriting the set of partitions above in terms of* X *and* Y *we have:*

$$\{\{X\}, \{\neg y\}, \{X \vee \neg y\}, \{X \wedge \neg y\}, \emptyset\}$$

3.1.1 (In)Discrete ProLT's

Motivation: As in point-set topology when considering the discrete topology we find that every possible combination of 'points' We can derive the discrete topology $\varsigma(D)$ from the combination of points in the unpartitioned basis $\{\{X\}, \{y\}, \{\neg X\}, \{\neg y\}\}$ in our partitioned basis $\{A, B, C, D\}$. This collection of vectors $\varsigma(\mathcal{D})$ satisfies the criteria for a topology as both $\varsigma(\mathcal{L})$ and $\emptyset$ are in the collection as well as both the unions and finite intersections of each set. We then have that $\varsigma(\mathcal{D})$ is indeed a topology, with all sets being open.

Definition *(Logical Presheaf on Open Sets)*
Let $\varsigma(\mathcal{L})$ be a ProLT. A *logical presheaf* $\mathscr{F}$ of logical sets $\varsigma(\mathcal{L})$ consists of the data:

1. For every open set $\varsigma(\mathcal{U}) \subseteq \varsigma(\mathcal{L})$, a logical set $\mathscr{F}(\varsigma(\mathcal{U}))$ and
2. For every inclusion $\varsigma(\mathcal{V}) \subseteq \varsigma(\mathcal{U})$ of open subsets of $\varsigma(\mathcal{L})$, a morphism

of logical sets $\rho_{UV} : \mathscr{F}(\varsigma(\mathcal{U})) \to \mathscr{F}(\varsigma(\mathcal{V}))$

which is subject to the conditions:

a. $\mathscr{F}(\emptyset) = 0$,

b. ρ_{UU} is the identity map $\mathscr{F}(\varsigma(\mathcal{U})) \to \mathscr{F}(\varsigma(\mathcal{U}))$ and

c. If $\varsigma(\mathcal{W}) \subseteq \varsigma(\mathcal{V}) \subseteq \varsigma(\mathcal{U})$ are three open subsets, then $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$.♠

First, let $\mathcal{L} = \mathcal{L}_1 \vee \mathcal{L}_2 \vee \cdots \vee \mathcal{L}_n$ and $U, V, W \subseteq \{\varsigma(\mathcal{L}_1), \varsigma(\mathcal{L}_2), \cdots, \varsigma(\mathcal{L}_n)\}$ for logical sets $\varsigma(\mathcal{L}_1), \varsigma(\mathcal{L}_2), \cdots, \varsigma(\mathcal{L}_n) \subseteq \varsigma(\mathcal{L})$ where $\mathcal{L} \in \mathbf{LogEx}$. We can associate to each open subset $\varsigma(\mathcal{U}) \subseteq \varsigma(\mathcal{L})$ an open sentence $\mathscr{F}(\varsigma(\mathcal{U}))$ which for every inclusion $\varsigma(\mathcal{V}) \subseteq \varsigma(\mathcal{U})$ of open subsets of $\varsigma(\mathcal{L})$ a restriction morphism of logical sets $\rho_{UV} : \mathscr{F}(\varsigma(\mathcal{U})) \to \mathscr{F}(\varsigma(\mathcal{V}))$ by letting the restriction morphism ρ_{UV} be that of the intersection $\{\varsigma(\mathcal{U})\} \cap \{\varsigma(\mathcal{V})\} = \{\varsigma(\mathcal{V})\}$ then we have a restriction morphism isomorphic to that of the restriction $(\mathcal{U} \vee \mathcal{L}_i) \mid_{\mathcal{V}} = (\mathcal{U} \vee \mathcal{L}_i) \wedge \mathcal{V} = \mathcal{V}$ where $\mathcal{L}_i$ is a subexpression of $\mathcal{L}$. We can see then that $\varsigma(\mathcal{U} \vee \mathcal{L}_i) \mid_{\varsigma\mathcal{V}} = \varsigma\mathcal{V}$ So our restriction morphism is equivalent to AND'ing a logical expression (i.e. $\mathcal{U}$) with another logical expression (i.e. $\mathcal{V}$) to which it is to be restricted by. The fcomposition of restriction morphisms is satisfied as well since, as shown above for an arbitrary choices of W, U, and V the restriction morphisms ρ_{UV} and ρ_{VW} hold and as $W \subseteq V \subseteq U$ we have that $W \subseteq U$ hence $\rho_{UV} \circ \rho_{VW} = \rho_{UW}$ ∎ Next we investigate how concepts as p-adic balls relate to topologies over logical expressions. In a simplistic overview; as p-adic balls have a unique representation in sets of natural numbers and, as we demonstrated previously that we can partition a covering set and relate it to a ProLT, we can find a correspondence between concepts and a ProLT. Our aim is to create larger complexes of concepts by gluing together smaller sets of concepts.

To begin, one way in which we can relate a concept to a logical expression is by the use of Godel Number. We are only interested in whether a concept could be uniquely related to a logical expression, after which we can create a ProLT from this expression. A theorem by Godel states that given a well formed formula in a formal language (such as propositional logic) there is a unique natural number which can be associated to this formula. When dealing with concepts we can convert each concept in to a logical expression of conjunctions and disjunctions of truth statements concerning the attributes of a concept.

As an overview that a unique Godel number can be found for any distinct concept consider the concept $U_{\frac{1}{2^2}} = U = \{\alpha_0 = \text{young}, \alpha_1 = \text{male}\}$ we could create the logical expression over U, call it $\mathcal{L}_U$ in which $\mathcal{L}_U = \text{young} \wedge \text{male}$. Thus this 2nd order concept can be represented by the conjunction of two propositions in which $\alpha_0 = T$ and $\alpha_1 = T$. This is of course different than the disjunction of two 1st order concepts $V_{\frac{1}{2^1}} = V = \{\beta_0 = \text{young}\}$ and $W_{\frac{1}{2^1}} = W = \{\gamma_0 = \text{male}\}$ where $V \vee W$ means being young or male. However, as each concept, in the conscious, is isomorphic to the powerset of subsets of its attributes (which we then create the discrete topology on), we have that U can be distinguished to be V or W, but in the unconscious $U = V \wedge W$. But as V and W are both well formed propositions in propositional logic, so is U. Thus we can associate a unique Godel number to each proposition.

The method for assigning Godel numbers just described shows the ability for the representation of concepts as natural numbers. For simplicity however, since we are only demonstrating algebraic topological ideas on small ordered concepts, we will just use prime numbers which is the simplified version of Godel's approach himself. We start by looking at the two concepts illustrated above, however instead of denoting attributes by $\alpha_0, \alpha_1, \cdots, \alpha_n$ we let each attribute α_j be denoted by truth variable such as W, X, Y, Z (as we shall see below) this way we can succinctly evaluate each of the attributes to find the corresponding truth valuation i.e. J or $\neg J$.

This begins to look like our representation of logical expressions above. It is important to note that because we are using 2-adic notation we have only 2 choices for each attribute i.e. young or old which we can re-phrase in respect to true or false in our representation. Of course as each integer can be represented as a sum of a 2-adic base expansion we can, once again, find a unique representation for a p-adic representation of attributes, that is; in which there are p possibilities for each attribute. Also we do not have to have possible values for attributes being opposites such as if we were concerned with only two colors $A = \text{red}$ and $\neg A = \text{green}$ we can interpret this as the only possible colors are red and green, and if one of the two choices is measured (distinguished), and it is not red, then it must be green.

We start by letting $W = \text{old}, \neg W = \text{young}$, $X = \text{man}, \neg X = \text{woman}$, $Y = \text{long hair}, \neg Y = \text{short hair}$, $Z = \text{large}, \neg Z = \text{small}$ and let us examine

a few possibilities for concepts and letting:

$$A = \{X, \neg Y, \neg Z\}, B = \{Y, \neg X, \neg Z\}, C = \{Z, \neg X, \neg Y\}$$

Now as mentioned before, the unconscious can distinguish between concepts, but does not see the sub-concepts of each concept. However the conscious can. Each concept has the discrete topology associated to it, hence, for example concept B has the discrete topology

$$\mathcal{D}_B = \{\{Y, \neg X, \neg Z\}, \{Y, \neg X\}, \{Y, \neg Z\}, \{\neg X, \neg Z\}, \{Y\}, \{\neg X\}, \{\neg Z\}, \emptyset\}$$

, which intersected with

$$\mathcal{D}_C = \{\{\neg Y, \neg X, Z\}, \{\neg Y, \neg X\}, \{\neg Y, Z\}, \{\neg X, Z\}, \{\neg Y\}, \{\neg X\}, \{Z\}, \emptyset\}$$

gives the subspace (relative) topology of $\mathcal{D}_B \cap \mathcal{D}_C$ or, for intuition (and gluing associated sections) $\mathcal{D}_B|_{\mathcal{D}_C} = \mathcal{D}_C|_{\mathcal{D}_B}$ which is $\{\{\neg X\}, \emptyset\}$ which is a topological space on an element of a first order category which was distinguished by the conscious.

We can see that X and $\neg X$ by themselves both represent elements of a first order category - those being the category of a man and that of a woman. We would now like to build a higher order category, that of say a person. We do this by gluing, that is, we look for compatible sections on presheaves of the data associated to each concept in which to glue together to create sheaves or higher order concepts. This will be the method of how we construct higher order simplicial complexes from smaller ordered ones.

3.2 Logical Complexes

Motivation:

3.2.1 Logical Ideals

Definition *(Logic Ideals)*

A *logical ideal* I is defined such that $\forall i \in I$

The ideals over a set of elements is the set generated over the operations $\circledvee$ and $\circledwedge$ between each pair of elements. The ideal I generated by $\langle[\wedge],[\vee],[\Updownarrow]\rangle = \{[\wedge],[\vee],[\Updownarrow],[0]\}$. If however we add $[\neg\vee]$ in the generators we get

$$\langle[\wedge],[\vee],[\Updownarrow],[\neg\vee]\rangle = \{[\wedge],[\neg\wedge],[\vee],[\neg\vee],[\Updownarrow],[\Leftrightarrow],[\circledcirc],[0]\}$$

The first six elements (the non-identity elements over $\circledvee$ $\circledwedge$ are exactly those elements which make up the larger 8-square as seen below. It is interesting to note that it is impossible to build an exact sequence (and thus an logical 4 element square) out of the four unique non-identity elements in the four element generated I. This can be seen as there is only a single element, that being $[\neg\vee]$ with the lower right element being a 1 and so there is no image to which it is the kernel.

We also have that this same ideal is generated by the 4-element set using the logical relationship $\Leftrightarrow$. This gives the analogous exact 8-square over the 6 unique non-identity unique elements

3.2.2 Exact Logical Sequence

Motivation:

There are some interesting phenomenon between logical operators, this becomes more apparent when looked at exact sequences. Exact sequences will eventually be used to help in building simplicial complexes.

Definition *(Exact Sequence)*

♠

Now we investigate exact sequences of logical expressions. Let $L_1, L_2 \in \mathcal{L}$ where $\mathcal{L}$ is the set of logical expressions. We build the short exact sequence:

$$0 \to L_1 \vee L_2 \to L_1 \wedge L_2 \to (L_1 \vee L_2)/(L_1 \wedge L_2) \to 0$$

As $(L_1 \vee L_2)/(L_1 \wedge L_2) = (L_1 \vee L_2) \wedge \neg(L_1 \wedge L_2) = (L_1 \Updownarrow L_2)$ we have found two expressions that are distinguishable from one another.

We use L_1 and L_2 as generalized logical expressions as we would like to move from the case of just two logical variables i.e. X and Y.

Example 3.11 *Let* $L_1 = \neg X \wedge Z$ *and* $L_2 = Y$ *Thus the expression* $(L_1 \vee L_2)/(L_1 \wedge L_2) = (X \wedge Z) \Updownarrow Y \Updownarrow Z$

Example 3.12 *Consider the exact sequence:*

$$0 \xrightarrow{\alpha} X \wedge Y \xrightarrow{\beta} X \vee Y \xrightarrow{\gamma} X \Updownarrow Y \xrightarrow{\delta} 0$$

Corresponding to the representation using the partitioned 3-atom Venn diagram **??***:*

$$0 \xrightarrow{\alpha} \{D, G\} \xrightarrow{\beta} \{A, B, D, G, E, F\} \xrightarrow{\gamma} \{A, B, E, F\} \xrightarrow{\delta} 0$$

and examining the exactness we see:

$im(\alpha) = 0 = ker(\beta)$

$im(\beta) = \{D, G\} = ker(\gamma)$

$im(\gamma) = \{A, B, E, F\} = ker(\delta)$ ♦

Consider the exact sequences of logic

$$0 \xrightarrow{\alpha} X \wedge Y \xrightarrow{\beta} X \vee Y \xrightarrow{\gamma} X \vee Y/X \wedge Y \xrightarrow{\delta} 0$$

Using the correspondence matrices we can find the quotient of one logical expression by another by simple matrix subtraction (however a method will be shown below that will be equivalent). The above exact sequence in matrix form is:

$$0 \xrightarrow{\alpha} [\wedge] \xrightarrow{\beta} [\vee] \xrightarrow{\gamma} [\vee]/[\wedge] \xrightarrow{\delta} 0$$

where $[\wedge]/[\vee] = [\Updownarrow]$. For this chain to be exact:

1. $im(\alpha) = ker(\beta)$ as α is the zero-map and β is injective.
2. $im(\beta) = ker(\gamma)$ as by construction $ker(\gamma) = [\vee]/[\wedge]$
3. $im(\gamma) = ker(\delta)$ as $im(\gamma) = [\Updownarrow] = ker(\delta)$

Also considering another exact sequence:

$$0 \xrightarrow{\alpha} [\Leftrightarrow] \xrightarrow{\beta} [⟴] \xrightarrow{\gamma} [⟴]/[\Leftrightarrow] \xrightarrow{\delta} 0$$

we can find the relationship between $\Updownarrow$ and $\Leftrightarrow$ using the splitting lemma. Note $[⟴]/[\Leftrightarrow] = [\Updownarrow]$

Definition *(Splitting Lemma)* ♠

Thus we end up with:

$$[\circledast] \simeq im(\beta) \oplus im(\gamma)$$

Note

Add method found of tensoring together logical expressions to find the quotient (Bible)!

W	X	Y	Z	Rep.	W	X	Y	Z	Rep.
W	X	Y	Z	A	$\neg W$	X	Y	$\neg Z$	I
$\neg W$	X	Y	Z	B	W	$\neg X$	Y	$\neg Z$	J
W	$\neg X$	Y	Z	C	$\neg W$	X	$\neg Y$	Z	K
W	X	Y	$\neg Z$	D	$\neg W$	X	$\neg Y$	$\neg Z$	L
W	X	$\neg Y$	Z	E	$\neg W$	$\neg X$	Y	$\neg Z$	M
$\neg W$	$\neg X$	Y	Z	F	$\neg W$	$\neg X$	$\neg Y$	Z	N
W	$\neg X$	$\neg Y$	Z	G	W	$\neg X$	$\neg Y$	$\neg Z$	O
W	X	$\neg Y$	$\neg Z$	H	$\neg W$	$\neg X$	$\neg Y$	$\neg Z$	P

Since we want to work in the a partitioned space of 4-logical sets W, X, Y, and Z we will represent, as was done in the case for the partitions of 3-logical sets (**??**), logical expressions by the disjunction of their respective partitions. This will be extremely useful in the upcoming computations. Here is the corresponding table:

Atom	Disjunctive Set
W	$\{A, C, D, E, G, H, J, O\}$
X	$\{A, B, D, E, H, I, K, L\}$
Y	$\{A, B, C, D, F, I, J, M\}$
Z	$\{A, B, C, E, F, G, K, N\}$

Definition *(Homotopic)*

Two projective resolutions are said to be *homotopic* if the morphisms are commutative. ♠

Example 3.13 *Consider the resolutions:*

$$0 \to X \wedge (Y \Updownarrow Z) \overset{d_3}{\to} X \wedge Y \wedge \neg Z \vee \neg Y \wedge Z \overset{d_2}{\to} \neg X \wedge (Y \vee Z) \overset{d_1}{\to} X \vee Y \overset{d_0}{\to} X \to 0$$

and:

$$0 \to X \wedge \neg Y \wedge Z \overset{d'_3}{\to} \neg Y \wedge Z \overset{d'_2}{\to} \neg X \wedge Z \overset{d'_1}{\to} X \vee (Y \wedge Z) \overset{d'_0}{\to} X \to 0$$

Which we can see the projectivity, when the resolutions are written in their partitioned form:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \{D,E\} & \xrightarrow{d_3} & \{C,D,E\} & \xrightarrow{d_2} & \{B,C,I\} \\
 & & \downarrow{\scriptstyle f_3} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_1} \\
0 & \longrightarrow & \{D\} & \xrightarrow{d'_3} & \{C,D\} & \xrightarrow{d'_2} & \{C,I\}
\end{array}$$

$$\begin{array}{ccccc}
\{A,B,D,E,F,I\} & \xrightarrow{d_0} & \{A,D,E,F\} & \longrightarrow & 0 \\
\downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle f} & & \\
\{A,D,E,F,I\} & \xrightarrow{d'_0} & \{A,D,E,F\} & \longrightarrow & 0
\end{array}$$

As the maps between sets are linear in the partitioned basis we see that every element in $d'_i \circ f_i = f_{i-1} \circ d_i \; \forall i$ ♦

Let $C_1 = W \wedge X \wedge Y$ and $C_2 = X \wedge Y \wedge Z$ bet two concepts. In the unconscious, while C_1 is distinguishable from C_2, the unconscious can not see the structure of C_1 or C_2. Both the discrete and indiscrete topolgies are equivalent in the unconscious; that is $\mathcal{D}_{C_1} = \mathcal{I}_{C_1} = \{\{C_1\}, \emptyset\}$ and $\mathcal{D}_{C_2} = \mathcal{I}_{C_2}\{\{C_2\}, \emptyset\}$ and hence the subspace topology is $\emptyset$ as $\mathcal{D}_{C_2} \cap \mathcal{D}_{C_1} = \emptyset$. As there are only, as seen by the unconscious, two concepts, each in their own space and only related to themselves and the subspace topology on the two concepts is empty. The unconscious doesn't have a subspace of concepts.

However if we 'unfold' the concepts using the unfolding ∂ operator:

$\partial(C_1)$	$\partial(C_2)$
$\partial_W(C_1) = X \wedge Y \wedge \neg W$	$\partial_W(C_2) = 0$
$\partial_X(C_1) = W \wedge Y \wedge \neg X$	$\partial_X(C_2) = Y \wedge Z \wedge \neg X$
$\partial_Y(C_1) = W \wedge X \wedge \neg Y$	$\partial_Y(C_2) = X \wedge Z \wedge \neg Y$
$\partial_Z(C_1) = 0$	$\partial_Z(C_2) = X \wedge Y \wedge \neg Z$

Now as the ∂ operator unfolds concepts from the unconscious to the conscious we see that we can break apart larger concepts to smaller ones

One of the possibilities for the mind is to, upon measuring two different concepts, build a "higher dimensional space". This would mean that there is relation and comparison.

For example, consider the concept C = pink flower. While it would be constructed from the concepts C_1 = pink and C_2 = flower, hence C = pink $\wedge$ flower it might be the case that one of the sub-concepts, such as flower, is brought in to consciousness and then associated to the color pink, instead of the the other way around. Thus the flower is pink, whereas we could have the color pink and then associate a flower to it.

In either case; that of accessing the entire concept at once, or successively, we still have that on some level that we can build larger concepts from those smaller concepts e.g. the concept of a pink flower from the concept of the color pink and the concept of a flower. This is emphasized since, as we shall see, our methodology allows for an evolution of concepts from more simple to more complex. Hence when building a higher order concept C, consciousness sees 3 logical variables. Again it is important to note that whether the whole of a concept or only pieces can be seen simultaneously, if distinguishment happens there must be something to distinguish thus 'room' for another degree of freedom.

As an example, let $C_1 = \neg X \vee \neg Y)$ and $C_2 = \neg Y \vee \neg Z$ which in the 3-variable partioned basis then have that:

$$C_1 = \{A, B, C, E, F, H\} \text{ and } C_2 = \{A, B, C, D, E, H\}$$

Note that we are referring to C_1 and C_2 as concepts which have been unfolded in the conscious. While we aren't explicly mentioning, for example concerning $\partial(X \wedge Y)$, we can glue sections of the partition together to regain $X \wedge Y$. So when we refer to a concept viewed in the conscious as being unfolded keeping in mind that the conjunction of sub-concepts is not lost. Hence when we write the unfolded concept as viewed from the conscious, although it doesn't include the original concept unfolded (as a bi-product of the unfolding function), it can still be rebuilt. $\mathcal{D}_{C_1} = \{\{A, B, C, E, F, H\}, \emptyset\}$ and $\mathcal{D}_{C_2} = \{\{A, B, C, D, E, H\}, \emptyset\}$ Which gives us the subspace topology $\mathcal{D}_{C_1} \cap \mathcal{D}_{C_2} = \{\{A, B, C, E, H\}, \emptyset\}$. This is equal to the logical expression $Y \Rightarrow (\neg Y \wedge \neg Z)$. This means that if $\mathcal{V}(Y) = T$ then both $\mathcal{V}(X) = F$ and $\mathcal{V}(Z) = F$! We have

Y being distinguished from both X and Z whereas X and Z might not be able to be distinguished.

3.3 Logic Relationship

Definition *(Logical Function)*

A *logical relationship* between two logical matrices ar

Definition *(Logical Image and Kernel)*

The *logical image* of two

Definition *(Logical Triangle)*

A *logical triangle* is a triangle in which each of the points consists of a logical operator and pairing of edges between the logical operators is exact in both directions. ♠

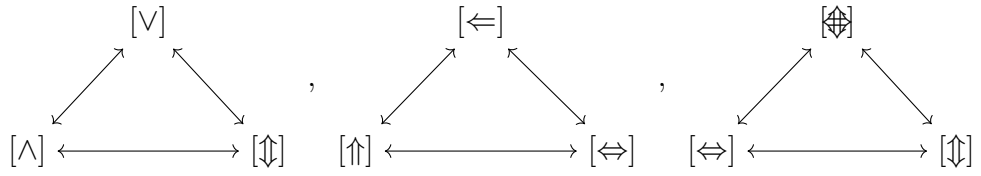
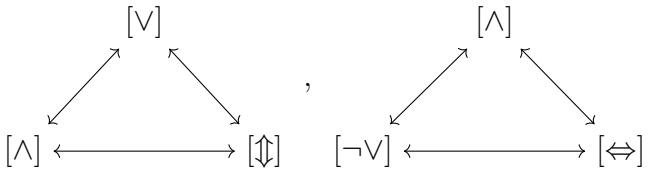


Figure 5: Logical triangles of logical operators

While, as one can see for example that $0 \to [\Leftrightarrow] \to [\nVdash] \to [\Updownarrow] \to 0$ is exact in both directions, the rotation $0 \to [\Updownarrow] \to [\Leftrightarrow] \to [\nVdash] \to 0$ is not.

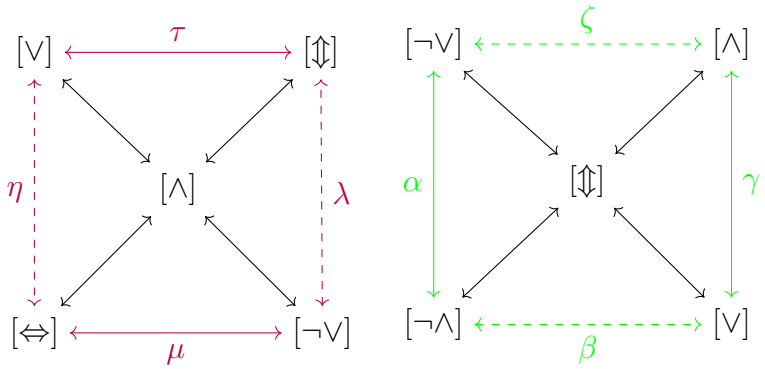
$\langle [\wedge], [\vee], [\Updownarrow] \rangle$ $\langle [\Leftarrow], [\Uparrow], [\Leftrightarrow] \rangle$ $\langle [\nVdash], [\Leftrightarrow], [\Updownarrow] \rangle$ The logical triangles are created from logical ideals of two elements which result in a set of 3 logical operations. For example $\langle [\wedge], [\vee], [\Updownarrow] \rangle$ and $\langle [\wedge], [\Leftrightarrow], [\neg\vee] \rangle$. Now as ideals are the sets of all logical matrices which, over the operators $\veebar$ and $\barwedge$ between all logical matrices, result in an a logical matrix also in the ideal, we must have that each logical matrix in a logical triangle is the XOR of the other two. As there may be more than one logical triangle with a particular logical matrix we are interested in finding out how we can build larger logical complexes from different logical triangles.

For example, consider the two logical triangles:



Both of the logical triangles have $[\wedge]$ in common. Let us attempt to 'glue' together these triangles over $[\wedge]$. In doing so we obtain the (left) square:

In both cases we have that the perimeter of the square is exact. This is

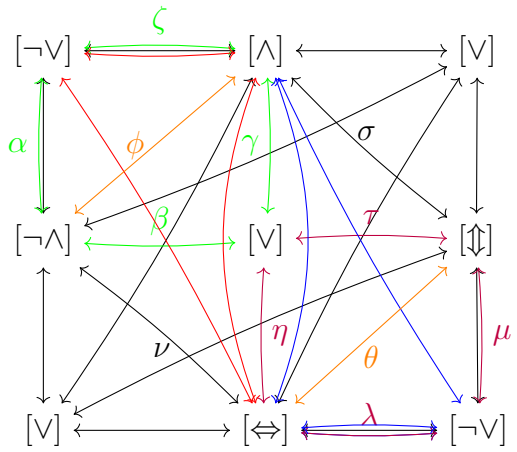


not always the case in that we can not create a logical square out of any two logical triangles. Note however that the logical matrices that are at the corners of the squares are negations of their diagonals. We also have that attempting to create logical triangles by combining logical matrices from the different logical triangles does not work either. This is denoted by a dotted line.

We do find some interesting relationships though, for example in the left (purple) square; first note that $\tau = [\vee]/[\wedge] = [⇳]$ and that $\mu = [\Leftrightarrow]/[\wedge] = [\neg\vee]$ While these values are expected due to the XOR (⇳) relationship between logical matrices of a logical triangle, we have that $[⇳]Ⓥ[\neg\vee] = [\neg\wedge]$ which is the XOR relationship between those logical matrices connected by the dashed line. More explicitly:

$$[⇳] ⇳ [\neg\wedge] = [\neg\wedge] = [\vee] ⇳ [\Leftrightarrow]$$

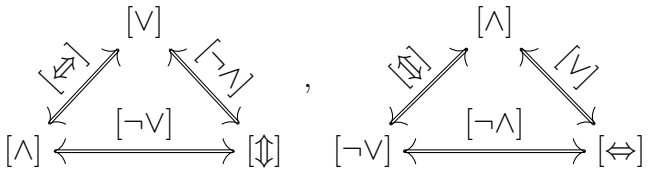
and hence there is a relationship between these logical triangles, a factor which we might glue over. We go a step further and join the two logical squares together at $[\vee]$ to get the following larger square:



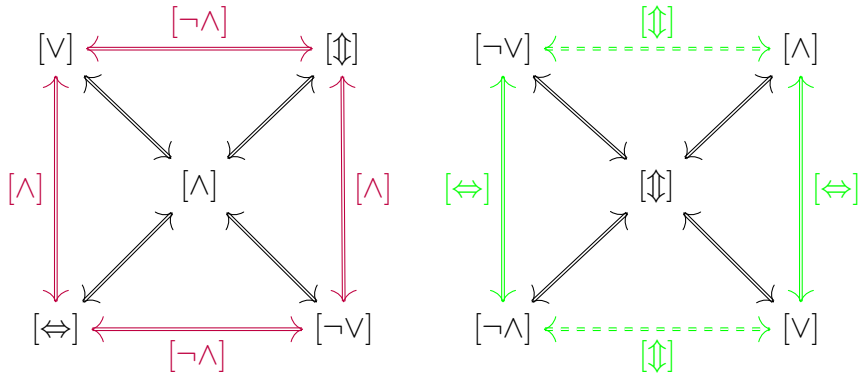
Now this is interesting because not only do we have a new exact 8-square we have that the unique elements of this 8-square are precisely the group over ⓥ and ⓐ generated by the ideal

$$\langle [\wedge], [\neg\wedge], [⇳] \rangle$$

Now consider the logical triangles created by connecting logical matrices by logical connectives, we have found an analog of the logical triangles which were exact and extended this 'exactness' to the ability to relate logical expressions together in a way which can be utilized by propositional logic. But furthermore we can build larger 'complexes' together with these logical triangles and build larger structures of logical statements! Consider the logical triangles:



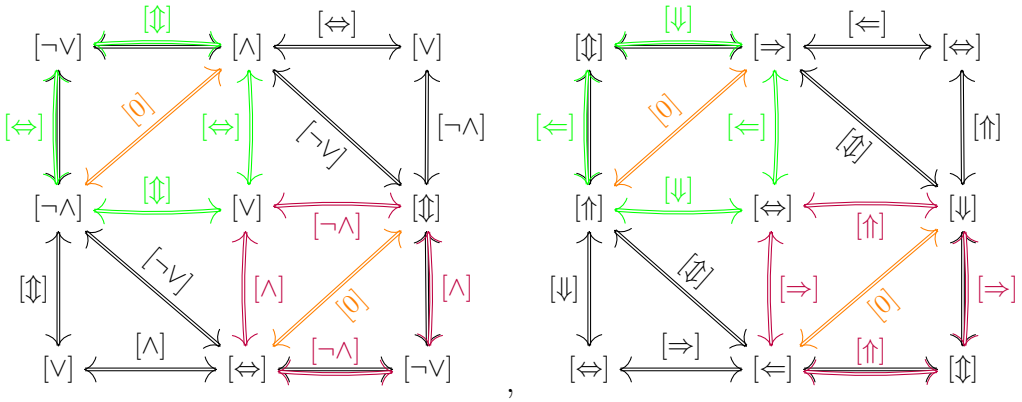
We can use these two triangles above to create a 4-square (or square) of logical expressions which are exact over the 'if and only if' mappings (⇔) as seen in the left (purple edged) square. The (green) square on the right is another built from the two logical triangles generated from the logical ideals $\langle [\vee], [\wedge], [⇳] \rangle$ and $\langle [\vee], [\neg\wedge], [\Leftrightarrow] \rangle$. Note that the ideals consists of elements



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which are generated from the others negation i.e. $[\vee] = \neg[\neg\vee]$ etc.

The next figure, the 8-square or square with 8 elements (not necessarily unique), on the left consists of the larger complex built by 'gluing' together the two 4-squares above (purple and green respectively). The 8-square on the right is added for completeness as it is shown that the logical matrices which it is built from don't have to be symmetric. It is actually this asymmetry that allows for the 'distinguishing' factor of the conscious and hence it is interesting to see how asymmetries could possibly occur from the symmetries after a possible 'collapse' of the logical wave function.



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3.4 Homological Algebra

Motivation: Homologies relate simplicial complexes via their boundaries and kernels. These bounaries (such as in point set topology) give a closure to the complexes.

Definition (*HomologyGroup*) The factor group $H_n(X) = Z_n(X)/B_n(X)$ is the n-dimensional *Homology Group* of X. ♠

Homology groups of a space are defined in terms of a triangulation and any triangulation of a space is isomorphic to any other triangulation; the Homology groups are the same so matter how the space is triangulated. This is important as it is an aim of this paper to give a relationship between different ideas as represented by simplices in which the morphisms between the simplices, whether or not they are continuous deformations, give information on the relationship between concepts.

Consider the homotopic exact sequences given in **??** we consider the homology. We now investigate exact sequences of the relationships between logical operators. We do this as we are interested in how 'different' one logical operator is from another. Consider the diagram: and we see that

$$\begin{array}{ccccccccc} [\Updownarrow] & \overset{\alpha_1}{\longleftrightarrow} & [\neg\wedge] & \overset{\alpha_2}{\longleftrightarrow} & [\Leftrightarrow] & \overset{\alpha_3}{\longleftrightarrow} & [\wedge] & \overset{\alpha_4}{\longleftrightarrow} & [\Updownarrow] \\ \Big\updownarrow \theta_1 & & \Big\updownarrow \theta_2 & & \Big\updownarrow \theta_3 & & \Big\updownarrow \theta_4 & & \Big\updownarrow \theta_5 \\ [\Updownarrow] & \overset{\beta_1}{\longleftrightarrow} & [\vee] & \overset{\beta_2}{\longleftrightarrow} & [\Leftrightarrow] & \overset{\beta_3}{\longleftrightarrow} & [\neg\vee] & \overset{\beta_4}{\longleftrightarrow} & [\Updownarrow] \end{array} \quad \begin{array}{c} (1) \\ \\ (2) \end{array}$$

this diagram is commutative and the relationships (over $\Leftrightarrow$) between logical matrices is as follows:

$$\beta_1 = [\neg\wedge] = \beta_3 \text{ and } \beta_2 = [\wedge] = \beta_4$$

$$\alpha_1 = [\vee] = \alpha_3 \text{ and } \alpha_2 = [\neg\vee] = \alpha_4$$

We can see that each of the operators α_i and β_j in (1) and (2) respectively are exact i.e. $\alpha_1 = \neg\alpha_2$.

We look at the Homotopy over these logical expressions, first note that:

$$\theta_1 = [\Diamond] = \theta_3 \text{ and } \theta_2 = [\Updownarrow] = \theta_4$$

which then we find that:

$$\beta_1 \circ \theta_1 = [\neg\wedge][\Leftrightarrow][\Diamond] = [\neg\wedge] = [\Updownarrow][\Leftrightarrow][\vee] = \theta_2 \circ \alpha_1$$

$$\beta_2 \circ \theta_2 = [\wedge][\Leftrightarrow][\Updownarrow] = [\neg\vee] = [\Diamond][\Leftrightarrow][\neg\vee] = \theta_3 \circ \alpha_2$$

$$\beta_3 \circ \theta_3 = [\neg\wedge][\Leftrightarrow][\Diamond] = [\neg\wedge] = [\Updownarrow][\Leftrightarrow][\vee] = \theta_4 \circ \alpha_3$$

$$\beta_4 \circ \theta_4 = [\wedge][\Leftrightarrow][\Updownarrow] = [\neg\vee] = [\Diamond][\Leftrightarrow][\neg\vee] = \theta_5 \circ \alpha_4$$

The image of $\alpha_1 \Leftrightarrow \alpha_2$ consists of those elements in α_1 which are true over $\Leftrightarrow$ that is; those elements in the pre-image α_1 which are the same as those in α_2. Those elements which are not the same then make $ker(\alpha_1)$ These elements are precisely $\neg\alpha_1$.

Looking at the commutative diagram we can see that we can define a homology over the two exact sequences (1) and (2). Explicitly

$$im(\alpha_1) = im(\alpha_3) = [\vee] = \neg[\neg\vee] = ker(\alpha_2) = ker(\alpha_4)$$

and

$$im(\beta_1) = im(\beta_3) = [\neg\wedge] = \neg[\wedge] = ker(\beta_2) = ker(\beta_4)$$

Thus we have that the homology of the modules (logical matrices) follows $im(\phi_i)/ker(\phi_{i+1})$ where ϕ is substituted with α and β and as, for example:

$$[\wedge]/[\neg\wedge] = [\wedge]$$

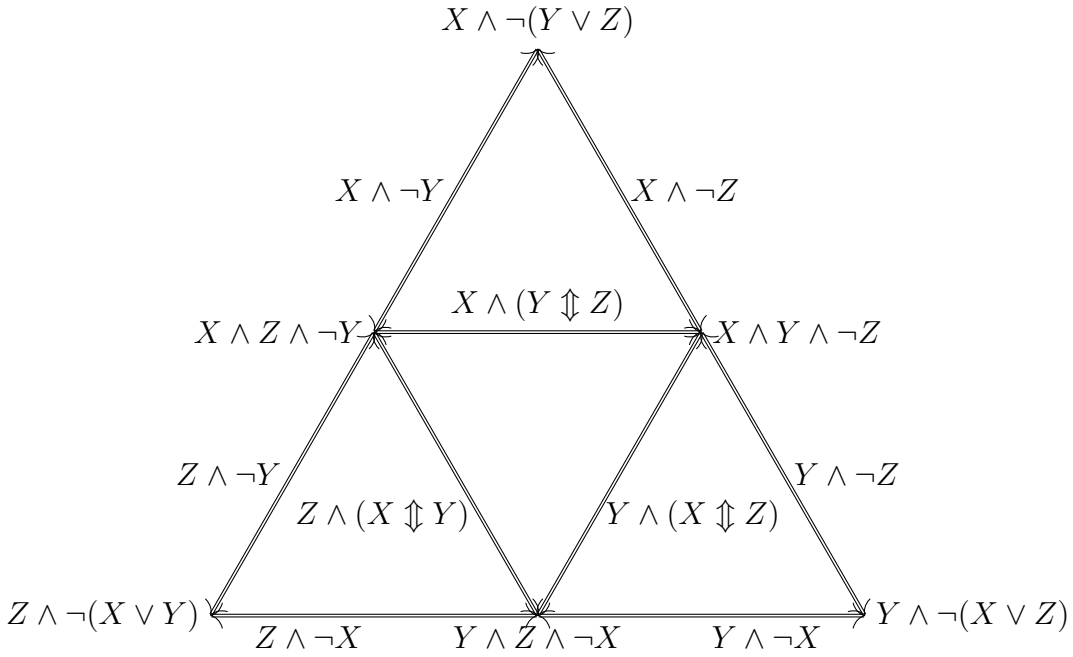
This is similar to group homology.

$$[\vee] \xrightarrow{\alpha_1} [\neg\wedge] \xrightarrow{\alpha_2} [\neg\vee] \xrightarrow{\alpha_3} [\neg\vee]$$

Say we wanted to "triangularize"

$$[\Leftrightarrow] \overset{\alpha}{\Leftrightarrow} [\Diamond] \overset{\beta}{\Leftrightarrow} [\vee]$$

We have that $\alpha = [\Leftrightarrow]$ and $\beta = [\vee]$ and $[\Leftrightarrow] \neq [\vee]$ but differ by $[\wedge$ as $[\vee] = [\Leftrightarrow]$



4 Conclusion

For **Logical Constructions**:

• Developed some logical structures such as the (Linear) Correspondence Matrices and was able to embed them in to more familiar Tensors and Bra-Ket notation so as to use in other areas of mathematics.

• Developed a way in which to view propositional logical expressions in terms of quantum mechanical principles.

• Found interesting operations such as matrix rotation which allow logical transformations. A possible algorithm for faster computations concerning propositional logical expressions?

For **Matte-Blanco's theory of the Mind**:

• Developed the notion of "Concepts" using simplical complexes which allows for the relation of information in a "Mental Space".

• Found a possible model for the "Unconscious" using Ultrametrics (Non-Archimedean Geometry) which allows for symmetries and indistinguishabil-

ity of points.

• Found a possible model for the "Conscious" using principles of quantum mechanics, more specifically by using superposition and the collapse of logical wavefunctions.

• Developed a logical operator, the "Unfolding" operator ∂ which is able to distinguish between literals of logical expressions.

For building a **"Mental Space"**:

• Developed a logical topology as well as concepts such as logical discrete and indiscrete topologies which allow for overlapping of "Concepts" and possibly show the logical symmetry breaking when examining unconscious as compared to the conscious.

• Developed a number of structures in order to build this idea of a mental space which include logical analogs of exact sequences, ideals and presheaves.

• Build larger structures (complexes) using found logical structures, examples include logical triangles and squares of exact sequences of logical correspondence matrices.

• Employed algebraic topological principles such as homotopy and homolgy and showed that there are corresponding notions when using exact sequences of the found logical correspondence matrices.

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